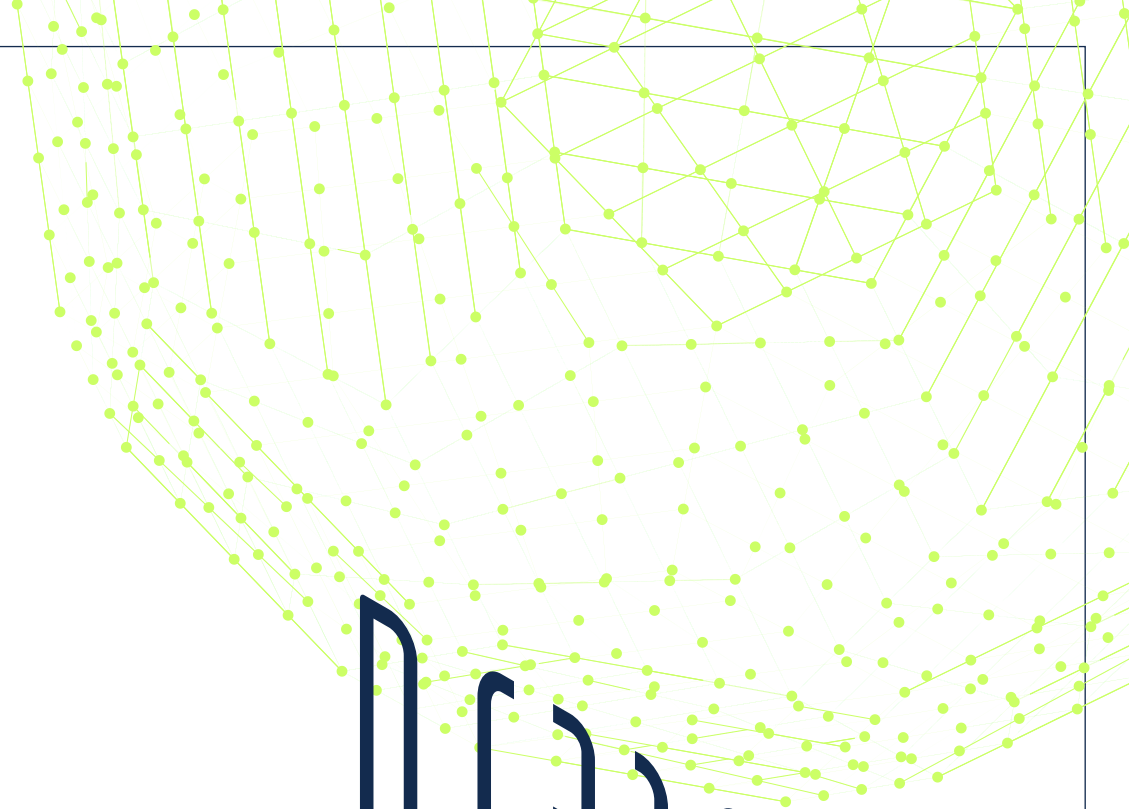
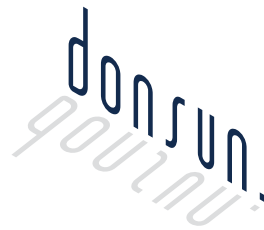




DonSun Technical Manual

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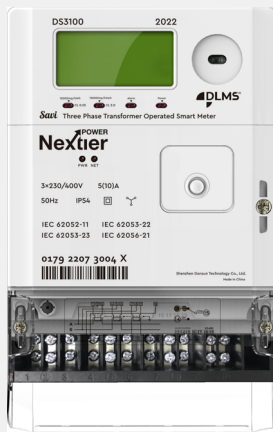
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Chapter 1

Introduction to DS3100

1.1

General Information



DS3100 *Savi*¹

Three Phase Transformer Operated Smart Meter

DS3100 *Savi* Three Phase Transformer Operated Smart Meter is manufactured by Donsun Technology. It provides an easily affordable solution for commercial and industrial smart metering. Robust, reliable, modern and equipped with innovative features. Supplemented to this is the ability for the meter to operate as a prepayment or post-payment meter by standard STS switching TOKEN.

DS3100 is compliant to IEC 62053-22, having an ingress protection of IP54 to IEC 60529 and comply with EMC standard IEC 50081-1.

DS3100 includes all the high functionality: four-quadrant metering; active, reactive, and apparent energy and demand measurements; standard event logs and time-of-use metering; LCD with optional backlight provides a flexible interface to the user for displaying meter data; and non-volatile memory for both billed and non-billed interval data.

The manufacturer of DS3100 *Savi* Three Phase Transformer Operated Smart Meter, Donsun Technology appreciates your choice on us. As a professional and agile technical team, we are devoted to providing the most optimal products and solution to utilities, partners and most importantly the customers.

Our core business is to improve the revenue of our customers and improve the satisfaction of end users. Whatever you need, we are always beside you.

We hope you would find our product and services satisfactory.

“
ONE MODELLING
BIG CUSTOMER
METERING

¹ *Savi* is the registered mark of Steama Company Limited.

1.2

Brief Characteristics



STANDARDIZATION

DS3100 meter from Donsun Technology has been certified with STS (Standard Transfer Specification), taking the 20 digits STS TOKEN as the credit transfer. The communication protocol of the meter complies with DLMS requirements and has obtained the DLMS/COSEM certificate. Meanwhile the meter is designed and produced pursuant to IEC 62053-22, IEC 62055-31 and IEC 62056-21 as well as ISO 9001.



VERSATILITY

DS3100 offers various types of communication modules which is exchangeable and hot-swappable, including GPRS, PLC, WiSun, LoRa, NB-IoT and others. The Client can select the most suitable communication modules as per their actual condition and budget, which enables AMI/AMR implementation.



AVAILABILITY

DS3100 offers split prepayment utilization, which consists of Measurement Control Unit (MCU) and Customer Interface Unit (CIU). The communication between the MCU and the CIU can be twisted pair cable (M-Bus), RF, PLC or other designated method by Client.



FLEXIBILITY

DS3100 provides options of the credit reduction in both kWh and currency, namely STS or CTS. Client can choose the best option for their own utilization together with the option of Time-of-Use, which can configured via the communication ports of the meter as optical port, RS-485 locally or via remote communication issued from the Head End System.

1.3

Important User Information

DISCLAIMER

Shenzhen Donsun Technology Co., Ltd. ("DONSUN") (herein after refer as "Manufacturer") reserves the right to make changes in specifications and other information contained in this document without prior notice, and the reader should in all cases consult manufacture to determine whether any such changes have been made. The information in this publication does not represent a commitment on the part of manufacturer.

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The information contained in this manual is subject to modification without prior notice and does not impose any obligation on manufacturer. The software is used at owner's risk. Manufacturer cannot be held liable for damage arising from program errors or incorrect handling.

APPLICABILITY

The user manual contains all the information required for application of the meters for the intended purpose, which includes

- Provision of knowledge concerning characteristics, construction and function of the meter;
- Information about possible dangers, their consequences and measures to prevent any danger;
- Details concerning the performance of all work throughout the service life of the meter.

The contents of this user manual are intended for technically qualified personnel of energy supply companies responsible for the meter planning, installation, operation and maintenance of the meter.

1.4

Safety Signs

Signs and Symbols Used in this Technical Manual are compliant to IEC 82079-1.



Danger

Danger Indicates an imminently hazardous situation which will result in death or severe injury if not avoided.

A

Warning

Warning indicates a potentially hazardous situation which can result in death or severe injury if not avoided.

B**C**

Caution

CAUTION indicates a potentially hazardous situation which can result in minor or moderate injury if not avoided.

D

Note

NOTE indicates information content/Hints for proper, safe error free functionality.

1.5

Normative Standard

IEC 62052-11	Electricity metering equipment (a.c.)-General requirements, tests and test conditions- Part 11: Metering equipment
IEC 62053-22	Electricity metering equipment (a.c.) –Particular requirements –Part 22: Static meters for active energy(classes 0.2S and 0.5S)
IEC 62053-23	Electricity metering equipment (a.c.) – Particular requirements –Part 23: Static meters for reactive energy (classes 2 and 3)
IEC 62056-21	Electricity metering – Data exchange for meter reading, tariff and load control – Part 21: Direct local data exchange
IEC 62055-41	Electricity Metering-Payment systems Part41: Standard transfer specification (STS) Application layer protocol for one-way token carrier systems
IEC 62055-51	Electricity Metering-Payment systems-Part51: Standard transfer specification (STS)-Physical layer protocol for one-way numeric and magnetic card token carriers
IEC 62056-42	Electricity metering – Data exchange for meter reading, tariff and load control – Part 42: Physical layer services and procedures for connection-oriented asynchronous data exchange
IEC 62056-46	Electricity metering – Data exchange for meter reading, tariff and load control – Part 46: Data link layer using HDLC protocol
IEC 62056-47	Electricity metering – Data exchange for meter reading, tariff and load control – Part 47: COSEM transport layer for IP networks
IEC 62056-53	Electricity metering – Data exchange for meter reading, tariff and load control – Part 53: COSEM Application layer
IEC 62056-61	Electricity metering – Data exchange for meter reading, tariff and load control – Part 61: OBIS Object identification system
IEC 62056-62	Electricity metering – Data exchange for meter reading, tariff and load control – Part 62: Interface classes
IEC 60044-1	Part 1: Current Transformers, Instrument Transformers
IEC 62262-1	Degrees of Protection provided by Enclosures for electrical equipment against external mechanical impacts (IK Code)
IEC 60529:2001	Degrees of Protection provided by Enclosures (IP Code)

* Other homologation is available upon request.

1.6

Terminology & Acronyms

AC

Alternative Current

EMC

Electro-Magnetic Compatibility

EEPROMElectrically Erasable Programmable
Read Only Memory**ESD**

Electrostatic Discharge

I_n

Norminal Current

I_{max}

Maximum Current

I_{st}

Starting Current

LCD

Liquid Crystal Display

LED

Light Emitting Diode

MD

Maximum Demand

NVM

Non-volatile Memory

OBIS

Object Identification System

RTC

Real Time Clock

U_n

Norminal Voltage

ACRONYMS**A**

Ampere (Unit of currnent)

Hz

Hertz (Unit of frequency)

KG

Kilogram (Unit of weight)

KVAH

Kilo Volt Ampere Hour (Unit of apparent energy)

KVARH

Kilo Volt Ampere Reactive Hour (Unit of reactive energy)

KWH

Kilowatt-hour (Unit of active energy)

MM

Minimeter (Unit of length)

°C

Degree Celsius (Unit of Temperature)

V

Volt (Unit of voltage)

UNITS

1.7

Working Principle

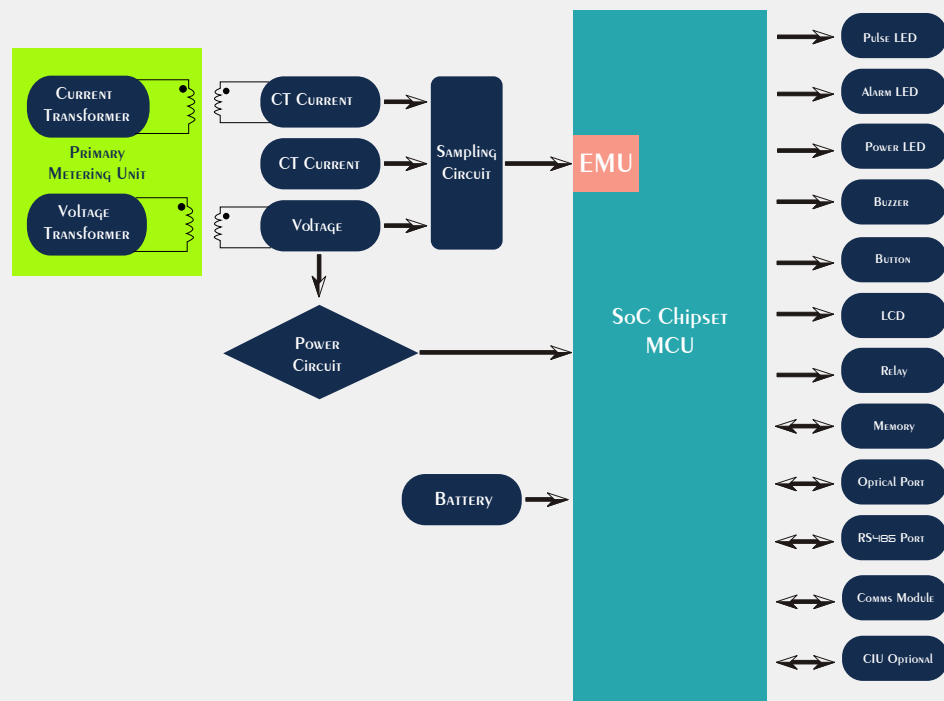
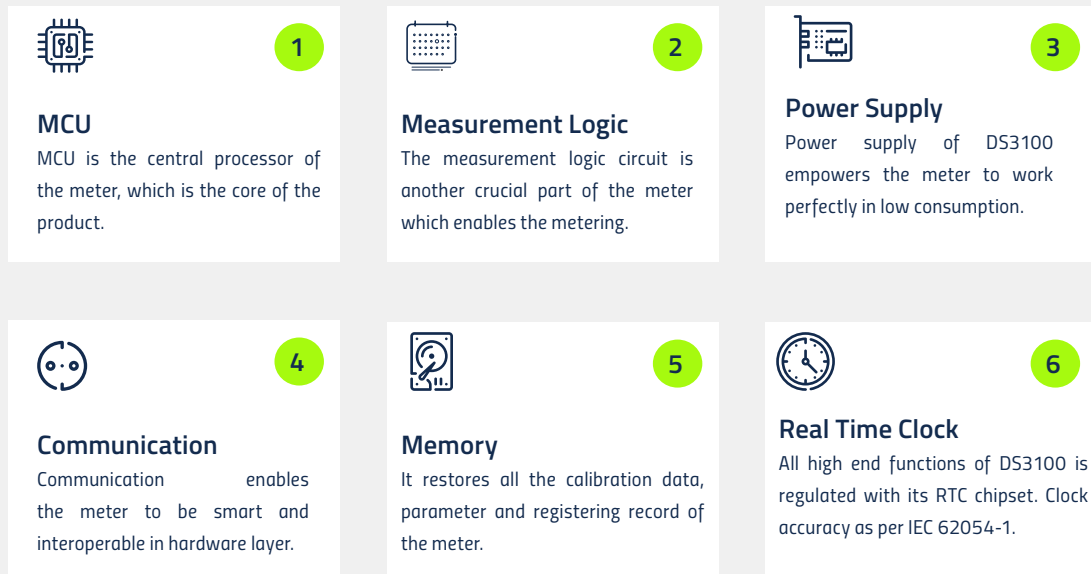


Figure 1 - Principle Diagram

- * DS3100 is powered by SoC solution with independent MCU for meter data and processing. It also includes the independent digital computation engine (CE) as well as FIR filter and ADC.
- Its peripheral includes LCD display, keypad and output for local communication including RS-485, optical and M-Bus for CIU communication. In addition, it provides various tamper detection mechanism for meter and terminal cover opening detection and breaker for disconnection at tamper occurrence.

Chapter 2 Measurement Control Unit

2.1 Specification Datasheet

Description	Specification
Reference voltage (U_n)	3X230/400V
Operation voltage	60% U_n ~140% U_n
Limit of Voltage	0% U_n ~190% U_n
Maximum switching voltage	>418V
Maximum withstand voltage	500V, 48 hours, no damages or accuracy deviation after restoration
Short-time withstand voltage	600V, 1 minute, no damages or accuracy deviation after restoration
Basic current (I_n)	5A
Maximum current (I_{max})	10A
Short-circuit current	3,000A, <3ms, no damage or accuracy deviation after restoration
Frequency	60Hz $\pm$ 5%
Accuracy class	Cl.0.5S for active energy, Cl.2 for reactive energy
Starting current	0.2% I_n (10mA)
Pulse constant	10000imp/kWh, 10000imp/kVarh
Connection type	3P4W
Consumption	Specification
Voltage circuit	<1W, 2.5VA @230V
Current circuit	<0.1VA @ Base current (Ib)
Environment	Specification
Operating Temperature	-25°C~70°C
Storage Temperature	-40°C~90°C
Humidity	average $\leq$ 95%, maximum $\leq$ 100%, non-condensing
Altitude	$\leq$ 3000m
EMC	Specification
Electrostatic discharge (ESD)	Contact discharge: 8kV Air discharge: 15kV, IEC 61000-4-2
Fast transient burst	4 kV/100 KHZ, IEC 61000-4-4
Immunity to HF Fields	80MHz-2GHz@10V/m with load; 80MHz-2GHz@30V/m with no load - accuracy not affected by magnetic fields from all sides -front, sides, top and bottom of the meter
Conducted disturbances by RF	150kHz - 80MHz 10V/m, IEC 61000-4-6
Radio Interference	CISPR 22, IEC 61000-4-2,3,4,6
Surge Immunity	30KA/KV , 4/10S against lightning, IEC 61000-4-5
Insulation	Specification
Impulse voltage (1.2/50 μ S to IEC 62052-11)	>8KV coil to contact; >4KV contact to contact
AC voltage (1MIN, 50HZ)	4KV coil to contact; 2KV contact to contact

2.2

Structural Overview Diagram

DS3100 takes an up to date yet classic structure design, with robust and compact meter structure, keypad, optical port, LCD, and other I/O ports. The PCBA of the meter is conformal coated to minimize the effect of water ingress, dust, vermin, chemicals and temperature extremes. The meter is designed to operate in harsh environment in tropical climate. DS3100 supports 2-phase 3-wire connection.

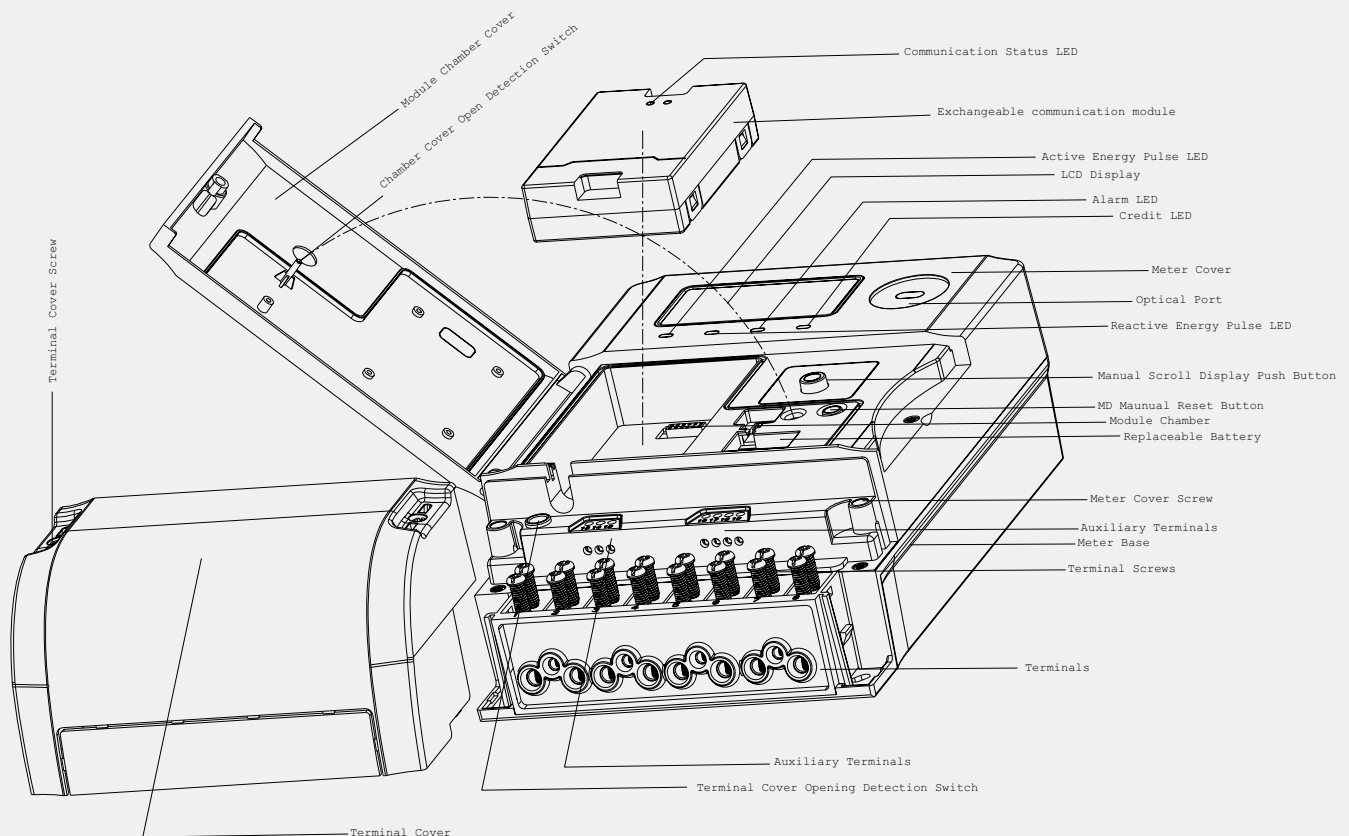
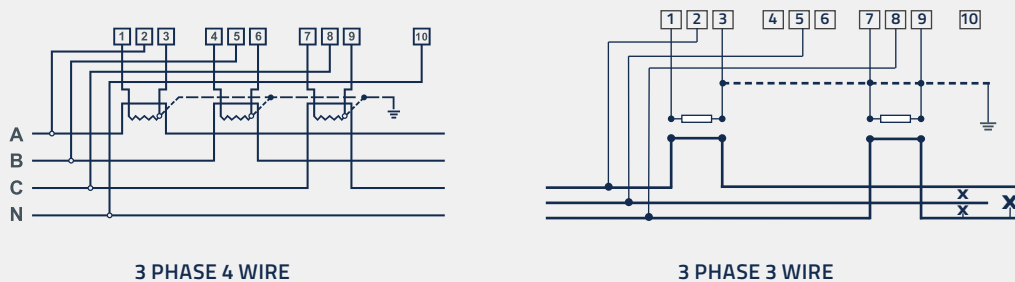


Figure 2 - Structural Diagram



STRUCTURAL STATISTICS



V0

Fire Resistance
IEC 60695-2-11



IP54

Ingression Protection
IEC 60529



2

Insulation Grade
IEC 61140:2016

2.3

Structural Description

DETAILED DESCRIPTION OF THE METER STRUCTURE

Meter Mounting

DS3100 is for front projection mounting, provided with 3 installation mounting points. The central hanging hole is on the upper side of the meter base and two lower mounting holes on the left side and right side of the terminal block under the terminal cover. The meter can be installed on any surface flat and meter box. Housing material has passed flaming retardant test; pass 960 degrees Celsius glow wire test IEC 60695-2-1. It is compliant to UL94-V0 rated @1.5mm with No toxic gases emitted.

Housing

Meter cover and terminal cover are made of UV resistant and long durable materials that the internal parts of the meter can not be accessed without removing the meter seals. The meter cover can not be opened unless removing the terminal cover. The meter housing is ultrasonically sealed between the base and meter cover against tampering with non-reversing fit once screws, meeting the requirement of IP54. The terminal cover of the meter has the knock-offs at the bottom for cable entry to facilitate the installation of the meter.

Terminals

The conductor terminals are front bottom access type, with the potential link placed inside the meter, made of nickel-plated H59 brass. The connection diagram is engraved inside the terminal cover. The meter terminal cover is made of transparent polycarbonate and in long extended type. The terminal can withstand continuous over current up to 125%I_{max}. The sealing of the terminal covers are sealed with our seals with terminal compartment reserved for utility sealing. Terminal holes and screws are made of nickel plated brass. Material of terminal block to pass tests of ISO 75.



MATERIAL

Polycarbonate Meter Cover
PC+20% GF Terminal Block
Flame retardant and 960°C glow
wire as per IEC 60695-2-1



SEALS

2 Seals for Meter Cover
1 Seal for Communication Chamber
2 Seals for Transparent Terminal Cover



TERMINAL COVER

Long Extended Transparent
Front Sealed



TERMINALS

Accommodates Soft and Hard
Drawn Cable, Brass Nickel-Plated



DISPLAY WINDOW

Display angle (≥120°), anti-ultraviolet
function, wide temperature range of
LCD display mode



MARKING

English, Indelible

DURABILITY STATISTICS



15

Years Service



15

Life-long Accuracy



6+2

Display Digits



85

Celsius Resistance

2.4 Dimension Diagram

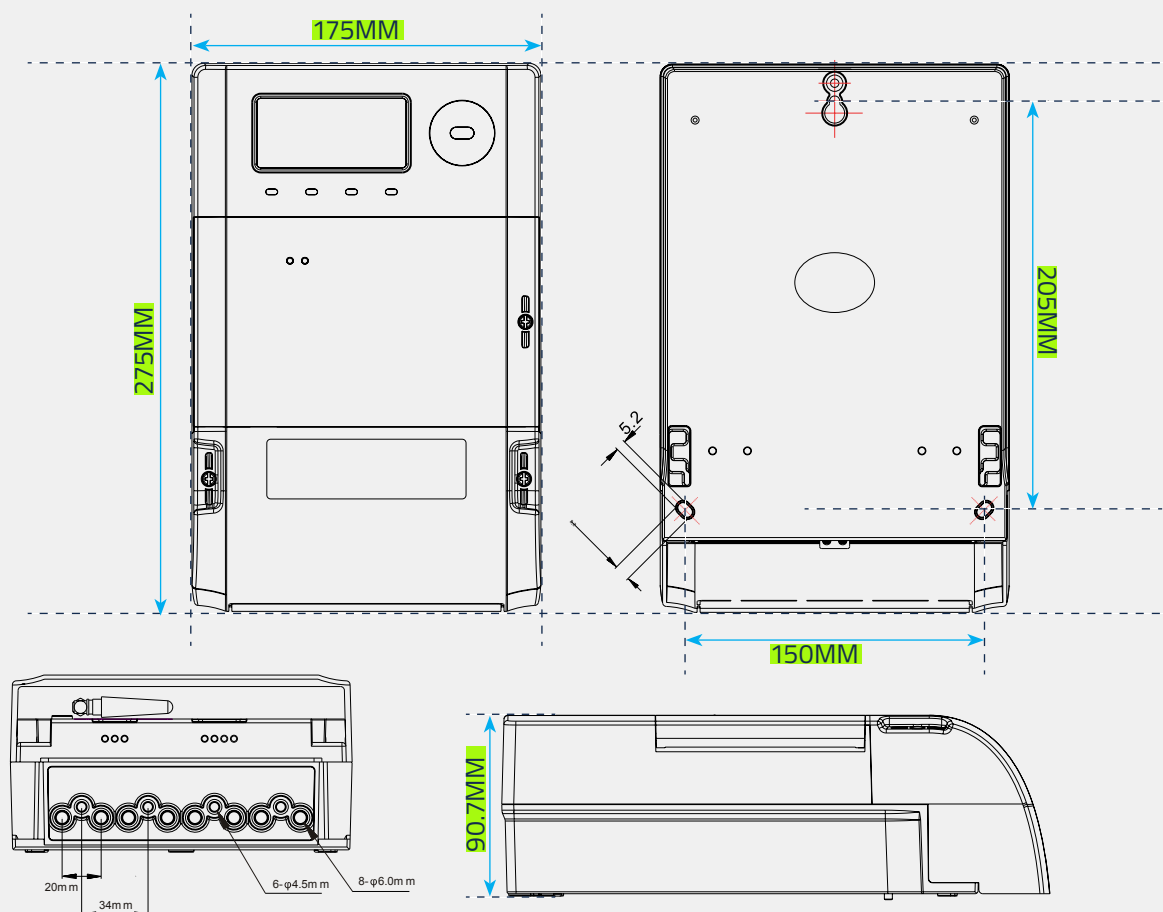


Figure 3 - Dimension Diagram

PHYSICAL DESCRIPTION

Furnished with a standard UV stabilized unbreakable high-grade flame retardant polycarbonate meter body enclosure, which complies with IEC 60695-2-11:2014. The terminal cover has the cable entry knock-offs to facilitate the installation.

TERMINAL CAPACITY

The current terminals of DS3100 are provided with two screws for each to ensure secure attachment of contactors. The terminals are made of H59 class brass with nickle-plated, complied to IEC 63004, to deliver sound electrical connectivity. Material of terminal block to pass tests of ISO 75.

INSTALLATION TOOLS

- * 1x6mm insulated terminal screw driver;
- * 1x insulated side cutters;
- * 1x electrical drill
- * 1x masonry drill bit - 6mm
- * Rawl plugs and screws

CABLING SPECIFICATION

Terminal diameter: 8mm
Terminal depth: 30mm
Hole distance: 20mm
Live: 10 mm²
Neutral: 10 mm²
Communication: 1 mm²
Supports 3P4W and 3P3W configurable by software.

2.5

Installation Process



Danger

INSTALLATION CAUTION

The installation and commission operator of energy meter must be technically qualified as below:

Trained with relevant technical training.

Be familiar with and comply with safety regulations, strictly comply with specific regulation of safety.

Please see if the required materials and tools are all prepared before starting operation.

TOOLS AND MATERIALS REQUIRED

The following tools and materials are necessary for installing the meter:

- The energy meter pre-set correctly (already opened an account) and sealed well.
- Customers' information chart of energy meter.
- Screws to fix energy meter.
- Seal and seal tools provided by the Utility.
- Slotted screwdriver and cross screwdriver which are used to secure screw.
- Drilling tools (if it is needed) and multi-meter.

01 //

MOUNTING THE METER

1. Position the meter between 1.3 and 1.5m above floor level, away from taps and not directly above the stove or sink (minimum distance-one meter).
2. Position and tape the template to the wall.
3. Drill holes through the template, deep enough to accommodate wall plugs.
4. Drill holes in wall base for cables / glands.
5. Fit the wall plugs and screw the wall base to the wall. Avoid over tightening as this can damage the wall base.

02 //

WIRING THE METER

1. Wire the meter using cable glands appropriate for the installation.
2. Connect the incoming supply to the top set of terminals (above the horizontal partition).
3. Connect the live wire to L line and the neutral wire to N line.
4. Connect all outgoing wires to the bottom set of terminals. Live to L load, neutral to N load.
5. The outgoing wires from the wall base should be connected to a ready board/ distribution board, if the earth leakage type meter is not being used.

03 //

COMMISSIONING AND SEALING

1. Apply power to the meter by closing the upstream breaker.
2. Check that the meter is displaying the available credit in kWh. If a zero balance is displayed, a credit token needs to be purchased.
3. Turn on an appliance or light in the consumer's premises, to ensure that the consumer has electricity.
4. Compare the meter number displayed, to the serial number printed on the meter label and supplied meter card, to confirm that the numbers match.
5. Seal the meter terminal cover with utility seals provided.
6. Handover the meter card and user guide the consumer.

2.6

Meter Installed with Distribution Cabinet

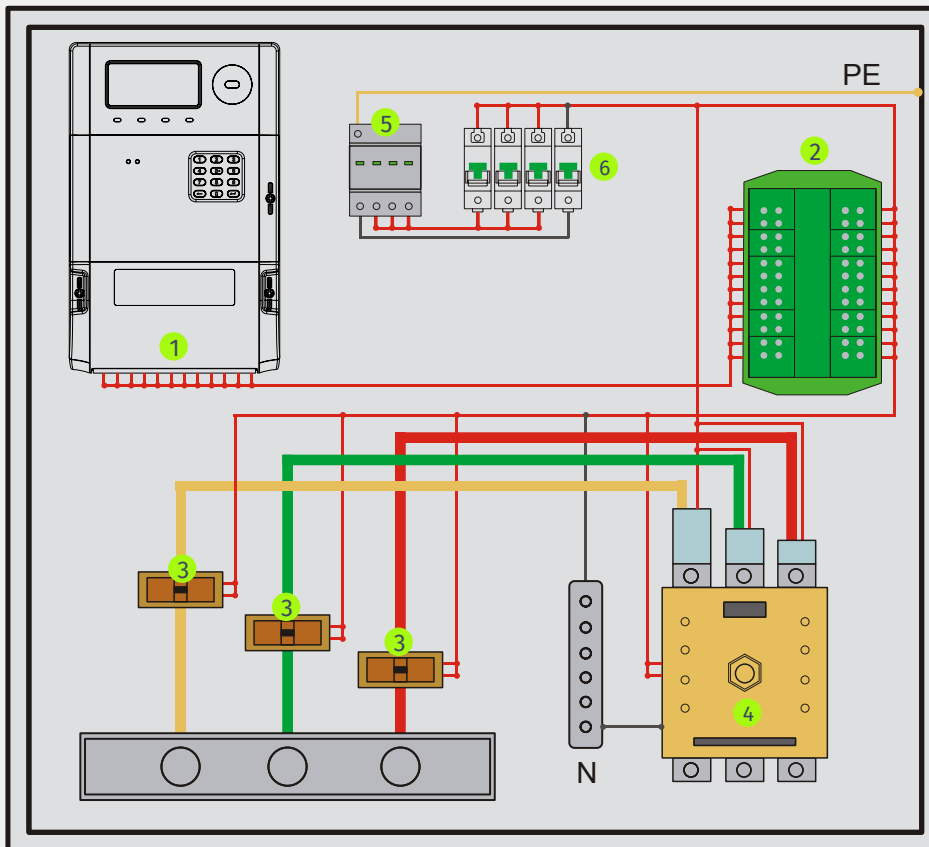


Figure 4 - Cabinet Installation Diagram



Meter

Meter takes the measurement and records data, which is the core of the product.



Terminals

The terminal board of the cabinet is to keep the wiring internal to be conformed to standards.



Transformer

Transformer externally installed empowers the meter to work perfectly in high consumption.



Circuit Breaker

Signalling from the meter, DS3000 can disconnect the load at ease.



Surge Arrester

Upto 45kV surge is immune by the metering cabinet, thanks to the surge arrester.



AC Contactor

Enabling the metering cabinet to connect to load and source safely.

Chapter 3 Functions

DS3100 includes all the high functionality: four-quadrant metering; active, reactive, and apparent energy and demand measurements; standard event logs and time-of-use metering; LCD with optional backlight provides a flexible interface to the user for displaying meter data; and non-volatile memory for both billed and non-billed interval data. In the event of a power failure, the meter secures all of the metering data. When power returns to the metering point, the contents of the secured metering data will be restored. DS3100 is built with an internal bi-stable latching relay to send to the MCCB for external load control.

3.1 Basic Features

- Active energy unit: kWh, with resolution as 0.01kWh;
- Reactive energy unit: kVarh, with resolution as 0.01kVarh;
- Apparent energy unit: kVah, with resolution as 0.01kVah;
- Display quantity: 999999.99 kWh/kVarh/kVah;
- When the display reaches to 999999.99 kWh/kVarh/kVah, the display will return to 0;
- Measures and accumulates forward and reverse energy separately into two different registers. Another total register for absolute value of import and export;
- Measurement and display of active energy in both import and export modes;
- The meter only only decrements credit in the forward direction.



- Energy unit: kWh, with resolution as 0.01kWh/kVarh/kVah;
- Display quantity: 999999.99 kWh/kVarh/kVah

MEASUREMENT RESOLUTION:

- Voltage: 0.01V
- Current: 0.001A
- Power: 0.001kW/0.001kvar/0.001kVA
- Power factor: 0.1%
- Frequency: 0.01Hz



- Power includes active power, reactive power and apparent power;
- Power factor is displayed in percentage;
- Indication of presence or absence of voltage of each phase

ENERGY MEASUREMENT

// 01

IMPORT & EXPORT KWH FOUR QUADRANT KVARH IMPORT & EXPORT KVAH

DS3100 measures and accumulates both import and export active, reactive and apparent energy into separate registers, and another total register for absolute value of import and export kWh/kVarh/kVah. The meter only decrements in the forward direction. Reverse energy is profiled in the export register separately.

RMS INSTRUMENTATION

// 02

VOLTAGE, CURRENT POWER, FREQUENCY, PF

DS3100 measures and displays instantaneous instrumental value as voltage, current, active power, reactive power, apparent power, power factor, phase angle and frequency.

3.1

Basic Features

MEASUREMENT RESOLUTION:

- Active MD unit: kW
- Apparent MD unit: KVA
- Active MD calculation can be configured to slide or block
- MD block is 15 min
- MD slide interval time is 1 min
- MD resolution: 0.001kW/ 0.001kvar/ 0.001kVA



MD will be transferred and reset as per defined billing schedule, after transfer the MD will be reset.

8 CHANNELS

- | | |
|--------------|----------------|
| - Time stamp | - Current |
| - Total kWh | - Power kW |
| - Demand kW | - Frequency |
| - Voltage | - Power factor |



The load profile mechanism works as follows:

- Programmable 1 minute to 60 minutes interval
- 180 days
- First In First Out (FIFO)

ODD HARMONICS

- Below the 15th harmonic of the Odd harmonic.
- The reference voltage is applied to the fundamental voltage
- Contents of the 3rd to 15th Harmonic Voltages: Apply 10% or less of the reference voltage.
- The phase angles of the fundamental and harmonic voltages must match at zero volts which is zero degree.
- Harmonic power factor: 1.0



THD is the instantaneous voltage reading for one second just before the storage time.

MAXIMUM DEMAND REGISTRY

// 03

ACTIVE & APPARENT KW/KVA SLIDE/BLOCK

DS3100 offers Maximum Demand in KW and KVA in different phase and TOU.

LOAD PROFILE

// 04

8 CHANNELS 30 MINUTES INTERVAL

DS3100 offers load profile for 180 days with interval of programmable 1 minute to 60 minutes in 8 channels.

TOTAL HARMONICS DISTORTION

// 05

15TH HARMONICS ODD HARMONICS

DS3100 offers odd harmonics up to 15th times of odd harmonics from fundamental voltage to 10% or less of the reference voltage.

3.1

Basic Features

BILLING

// 06

MONTHLY BILLING DAILY BILLING

The daily billing data includes Billing tag, total kWh (and tariffs), import kWh (and tariffs), export kWh (and tariffs), total kVarh (and tariffs), import kVarh (and tariffs), export kVarh (and tariffs), four-quadrant kVarh (and tariffs), total kVAh (and tariffs), import kVAh (and tariffs), export kVAh (and tariffs).

MONTHLY BILLING

- The meter records the monthly billing data for the latest past 12 months;
- The monthly billing transfer time can be configured, with default time at 00:00 of the first day of each month;
- Monthly billing data includes Billing tag, total kWh (and tariffs), import kWh (and tariffs), export kWh (and tariffs), total kVarh (and tariffs), import kVarh (and tariffs), export kVarh (and tariffs), four-quadrant kVarh (and tariffs), total kVAh (and tariffs), import kVAh (and tariffs), export kVAh (and tariffs), forward active MD and time of occurrence (and tariffs), reverse active MD and time of occurrence (and tariffs), forward reactive MD and time of occurrence (and tariffs), reverse reactive MD and time of occurrence (and tariffs).
- The meter records the monthly billing data for the latest past 12 months;
- The monthly billing transfer time can be configured, with default time at 00:00 of the first day of each month or defined by utility by PC software;

DAILY BILLING

- The meter records the monthly billing data for the latest past 31 days;
- The daily billing transfer time can be configured, with default time at 00:00 of each day or defined by utility by PC software;
- The meter's billing register can not be re-settable to zero readings;



- Monthly Import energy = quadrant 1 energy + quadrant 4 energy;
- Monthly Export energy = quadrant 2 energy + quadrant 3 energy;

TIME OF USE

// 07

SEASON TIMETABLE MONTH TIMETABLE

The ToU of DS3100 supports seasonal timetable, monthly timetable, as well as daily schedule.

TOU SCHEDULES

- Tariff number: 4
- Time zone: 12
- Daily schedule: 12
- Weekly schedule: 8
- Daily timetable: 8
- Exception days: 100



- ToU can be configured with PC software;
- Default tariff schedule when shipment is single tariff.

3.2

Other Features



1

Real Time Clock

Real-time clock quartz crystal oscillator controlled and with capabilities of resetting the clock without loss of billing data. It supports Remote and local clock synchronization, and back up with lithium battery, which has shelf life of more than 10 years.



2

Configuration

DS3100 supports configuration by optical port, or via TOKEN for different parameters.



3

Self-diagnosis

The meter can perform self-check on measurement error, Non-Volatile Memory (NVM) error, RTC error, latching relay abnormality, battery low voltage.



4

Anti-tampering

The meter detects and records meter terminal and meter cover openings, and reversed energy consumption.

Credit LED Status Indication (For prepayment)

 SUFFICIENT When the remaining credit is above the Level-1 pre-warning threshold, the LED stays up in Green.	 LOW CREDIT When the remaining credit is lower than the Level-1 pre-warning threshold, the LED flashes in Green.	 LOW CREDIT When the remaining credit is lower than the Level-2 pre-warning threshold, the LED flashes in RED in 1Hz.	 LOW CREDIT When the remaining credit is lower than the Level-3 pre-warning threshold, the LED flashes in RED in 2Hz.
 OVERDRAFT/ FRIENDLY MODE When the meter is in overdraft or friendly mode, the LED flashes in RED and Green alternately in 1Hz.	 NO CREDIT When the remaining balance is 0, the credit LED stays up in RED.	 CREDIT MODE When the meter operates in credit mode, the credit LED stays up in RED.	

LCD BACKLIT

The meter LCD backlight will be lit up when pressing any key of the meter.
The backlight would stay lit up when the meter enters lock screen mode.

Meter Data Reading Companion

- **PC Software:** The meter data reading PC software is provided with each batch of meter for the data reading and parameterisation and configuration. The meter software / programme is capable of tracking user access to the meter. Access to meter parameters and programming information shall only be through user-level password(s).
- **Optical probe:** Optical probes are provided for each batch of meter based on quantity of the meter.
- **Interrogator kit:** MC171 probes are provided for each batch of meter based on quantity of the meter.

3.3

Working Principle Descriptions

Meter Input

- Phase line L1,L2,L3, neutral line N, current I1, I2, I3

For the power supply to the meter

For sampling signal of measure

- Push buttons

For scrolling display

For order relay connecting and disconnecting manually

- External data signal inputs through communication interface

Meter Output

- LCD liquid crystal display with display buttons

- Electronic test impulse

- Signal outputs through communication interface

Metrology

Sample and calculate the input power grid signal to get related information, phase sequence has no affect on meter measurement, including following parameters:

- Active power (3 phases)

- Reactive power (3 phases)

- Voltage (3 phases)

- Current (3 phases)

- Frequency (one of phase with voltage)

- Power factor (3 phases)

- Harmonic voltage (3 phases) (total, 3rd to 21st odd harmonic)

- Harmonic current (3 phases) (total, 3rd to 21st odd harmonic)

Electric Test Pulse

- Active and Reactive power pulse is generated for testing the accuracy of the meter

3.4

Working Principle Descriptions

- The supply voltage for the meter is obtained from the power grid, ensuring the normal operation of internal parts of the meter.

Power Supply

- The battery supply connected in parallel with the normal supply ensures the operation of the meter free from interruption. When the normal supply is switched off, the backup battery has the capability to support the RTC in meter and detect cover opening tampering events.

Battery

- Real Time Clock is served as a time-base for calendar clock in the meter. It's mainly supported by the power supply when the grid power is on, but once interruption occurs, battery will ensure the normal operation of meter instead.

RTC

- fRAM

Memory

fRAM, for recording data which is used frequently, meter energy consumption is also memory in this fRAM, and is unchangeable.

- NVM

For meter has one piece of such memory, which is used for saving events records and lost record data; if needing upgrade, it is better to change more large memory for storing firmware update package.

- The connection and disconnection of user network can be controlled with relay, including load control and manual control.

Relay

- Optical

Communication Interface

Through this interface, master computers or HHU can communicate with meter with near-infrared communication.

- RS-485

For meter data local readout or meter parameterization, or through a universal bus for remote communication with an external communication module.

- Plug and Play Module

This part is situated in the meter, protected by the communication cover and terminal cover. It can be replaced by different communication module if there is necessary for upgrading to build an AMI system.

3.5

Meter Measurement

Analog Input

- Analogue signals include analogue voltage and current

Signal Conversion

- The AD converter in meter measuring system generates calibrated instantaneous digital values of voltage and current from the analogue input signals.

Data Preprocess

Signal processor determines the following digital mean values (averaged for one second in each case) from the instantaneous values and current generated by AD converter. Following are all mean values in 1 second.

- Active power (with sign for direction of power) (3 phases)
- Reactive power (with sign for direction of power) (3 phases)
- Apparent power (3 phases)
- Current (3 phases)
- Voltage (3 phases)
- Power factor (3 phases)
- Frequency
- Harmonic voltage(3 phases)(total, 3rd ~21st odd harmonic)
- Harmonic current(3 phases)(total, 3rd ~21st odd harmonic)

Data Process

The microprocessor calculates the following measured quantities from the mean values provided by the signal processor.

- Mean forward active power in 1 second (3 phases)
- Mean reverse active power in 1 second (3 phases)
- Mean reverse reactive power in 1 second (3 phases)
- Apparent power in 1 second (3 phases)
- Current in 1 second (3 phases)
- Power factor in 1 second (3 phases)
- Voltage in 1 second (3 phases)
- Power grid frequency

3.6

Signal Conversion & Processing

Signal Input Circuit

Input voltage signal is divided into low sampling voltages by high resistance voltage dividers of which resistance is $2.31\text{M}\Omega$ and 750Ω respectively. The proportional amount is 3080:1. For example, if the input signal U_2 is 230V, the sampling voltage U_{LX} will be 74.675mV and the passing current will be $99.567\mu\text{A}$. Then the apparent power consumption of the voltage sampling circuits can be calculated as 22.9mVA. The sampling current is obtained by using a current transformer placed in the meter. The ratio of current transformer is 1000:1, the sampling resistance of current transformer is 15Ω . For example, when applying an input current of 10A to the meter, the corresponding sampling voltage U_1 will be 15mV then apparent power consumption of current sampling circuits is less than 0.1VA.

Digitalizing

The input analogue signals is converted by AD converter in measuring system and then filtered. After that, the signals are calibrated and finally form the required digital instantaneous values.

Mean Value

From digital instantaneous values, measuring system calculates the mean values per second by integral calculus. Then the microprocessor scans these values at intervals of one second for further processing.

DATA PROCESSING

Signal Input Circuit

By scanning the mean values per second (active power, voltage, current, frequency, power factor), microprocessor forms related measured quantities. Meanwhile the active power per second is regarded as the active energy per second to be added into the total active energy consumption, the reactive power per second is regarded as the reactive energy per second to be added into the total reactive energy consumption, the apparent power per second is regarded as the apparent energy per second to be added into the total apparent energy consumption.

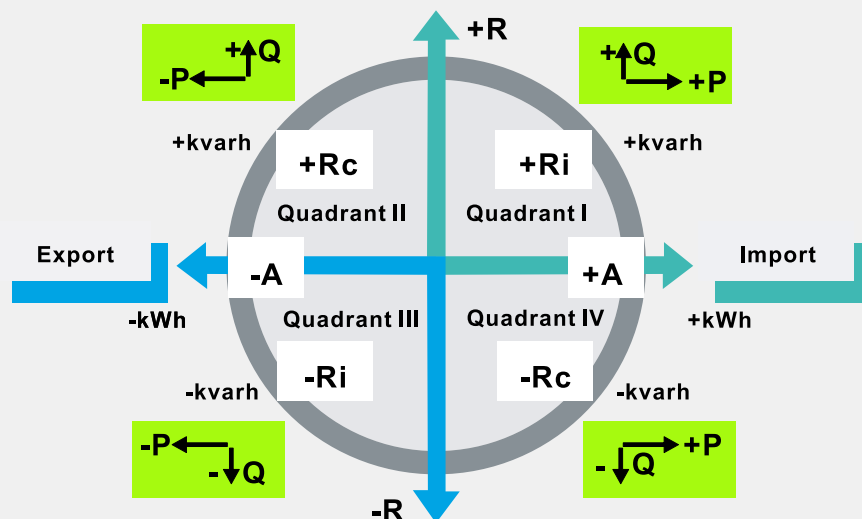


Figure 5 - Metrology Quadrants

3.7

Energy Accumulation

General

Microprocessor obtains the measuring data per second, after calculating, fifteen data are obtained: forward active power in 1s of three phases (namely forward active accumulation energy within 1s), reverse active power in 1s of three phases (namely reverse active accumulation energy within 1s), forward reactive power in 1s of three phases (namely forward reactive accumulation energy within 1s), reverse reactive power in 1s of three phases (namely reverse reactive accumulation energy within 1s), apparent power in 1s of three phases (namely apparent accumulation energy within 1s), above fifteen values are as input value of energy accumulation and are respectively accumulated into forward active total energy register, reverse active total energy register, forward reactive total energy register, reverse reactive total energy register, forward apparent total energy register, reverse apparent total energy register, quadrant I reactive total energy register, quadrant II reactive energy register, quadrant III reactive energy register, quadrant IV reactive energy register as well as corresponding tariff energy register according to current tariff.

Microprocessor then selects the information from the energy registers for display, load profile and remote readout.

Registers are included as follows:

Total Energy

- Total active energy register (TA)
- Total active energy tariff 1 register (TA1)
- Total active energy tariff 2 register (TA2)
- Total active energy tariff 3 register (TA3)
- Total active energy tariff 4 register (TA4)
- Total forward active energy register (+A)
- Total forward active energy tariff 1 register (+A1)
- Total forward active energy tariff 2 register (+A2)
- Harmonic voltage(3 phases)(total, 3 rd ~21 st odd harmonic)
- Harmonic current(3 phases) (total, 3 rd ~21 st odd harmonic)
- Total forward active energy tariff 3 register (+A3)
- Total forward active energy tariff 4 register (+A4)
- Total reverse active energy register (-A)
- Total reverse active energy tariff 1 register (-A1)
- Total reverse active energy tariff 2 register (-A2)
- Total reverse active energy tariff 3 register (-A3)
- Total reverse active energy tariff 4 register (-A4)
- Total forward reactive energy register (+R)
- Total forward reactive energy tariff 1 register (+R1)
- Total forward reactive energy tariff 2 register (+R2)
- Total forward reactive energy tariff 3 register (+R3)
- Total forward reactive energy tariff 4 register (+R4)
- Total reverse reactive energy register (-R)
- Total reverse reactive energy tariff 1 register (-R1)
- Total reverse reactive energy tariff 2 register (-R2)
- Total reverse reactive energy tariff 3 register (-R3)
- Total reverse reactive energy tariff 4 register (-R4)
- Total apparent energy register (TQ)
- Total apparent energy tariff 1 register (TQ1)
- Total apparent energy tariff 2 register (TQ2)
- Total apparent energy tariff 3 register (TQ3)
- Total apparent energy tariff 4 register (TQ4)
- Total forward apparent energy register (+Q)
- Total forward apparent energy tariff 1 register (+Q1)
- Total forward apparent energy tariff 2 register (+Q2)

- Forward active energy register of phase A (+Aa)
- Reverse active energy register of phase A (-Aa)
- Forward reactive energy register of phase A (+Ra)
- Reverse reactive energy register of phase A (-Ra)
- Forward apparent energy register of phase A (+Qa)
- Reverse apparent energy register of phase A (-Qa)
- Quadrant I reactive energy register of phase A (R1a)
- Quadrant II reactive energy register of phase A (R2a)
- Quadrant III reactive energy register of phase A (R3a)
- Quadrant IV reactive energy register of phase A (R4a)

Energy Phase A

- Forward active energy register of phase A (+Ab)
- Reverse active energy register of phase A (-Ab)
- Forward reactive energy register of phase A (+Rb)
- Reverse reactive energy register of phase A (-Rb)
- Forward apparent energy register of phase A (+Qb)
- Reverse apparent energy register of phase A (-Qb)
- Quadrant I reactive energy register of phase A (R1b)
- Quadrant II reactive energy register of phase A (R2b)
- Quadrant III reactive energy register of phase A (R3b)
- Quadrant IV reactive energy register of phase A (R4b)

Energy Phase B

- Forward active energy register of phase A (+Ac)
- Reverse active energy register of phase A (-Ac)
- Forward reactive energy register of phase A (+Rc)
- Reverse reactive energy register of phase A (-Rc)
- Forward apparent energy register of phase A (+Qc)
- Reverse apparent energy register of phase A (-Qc)
- Quadrant I reactive energy register of phase A (R1c)
- Quadrant II reactive energy register of phase A (R2c)
- Quadrant III reactive energy register of phase A (R3c)
- Quadrant IV reactive energy register of phase A (R4c)

Energy Phase C

3.8

Energy Calculation Explanation

Active & Apparent Energy

$$TA = |+A| + |-A|$$

|+A|: import active energy

|-A|: export active energy

$$TQ = |+Q| + |-Q|$$

|+Q|: import apparent energy

|-Q|: export apparent energy

Individual Phase Energy

Option 1: Algebraic accumulation: algebraic of three phase active, reactive and apparent energy in each second are as inputting data source;

Option 2: Absolute value accumulation: three phase energy is regarded as two independent single phase energy to accumulate, three phase energy always equal to sum of each phase.

Reactive Energy

$$+R = R1 + R4$$

$$-R = R2 + R3$$

R1: Quadrant I reactive energy

R4: Quadrant IV reactive energy

R2: Quadrant II reactive energy

R3: Quadrant III reactive energy

Individual Phase Energy

Option 1: Algebraic accumulation: algebraic of three phase active, reactive and apparent energy in each second are as inputting data source;

Option 2: Absolute value accumulation: three phase energy is regarded as two independent single phase energy to accumulate, three phase energy always equal to sum of each phase.

Total Power Factor

Current accumulation energy of each phase is the Algebraic accumulation

Total Power Factor= algebraic sum of each phase active energy in 1s/ algebraic sum of each phase apparent energy in 1s*100%

Current accumulation energy of each phase is Absolute Value Sum

Total Power Factor= absolute value sum of each phase active energy in 1s/ absolute value sum of each phase apparent energy in 1s*100%

3.9 Energy Calculation Explanation

Definition

Current demand average value: when it is still within current sliding window, this value is calculated every minute using this expression: (accumulating energy within current window/ numbers of period*period)

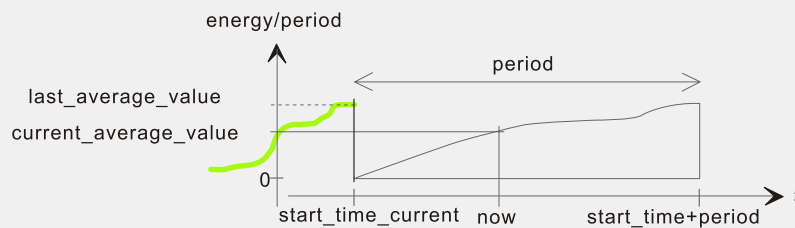
Final demand average value: after an integrating period, this value is calculated every integrating period using this expression: (accumulating energy within current window/ numbers of period*period)

MD in current month: Max value of final demand average value in current month

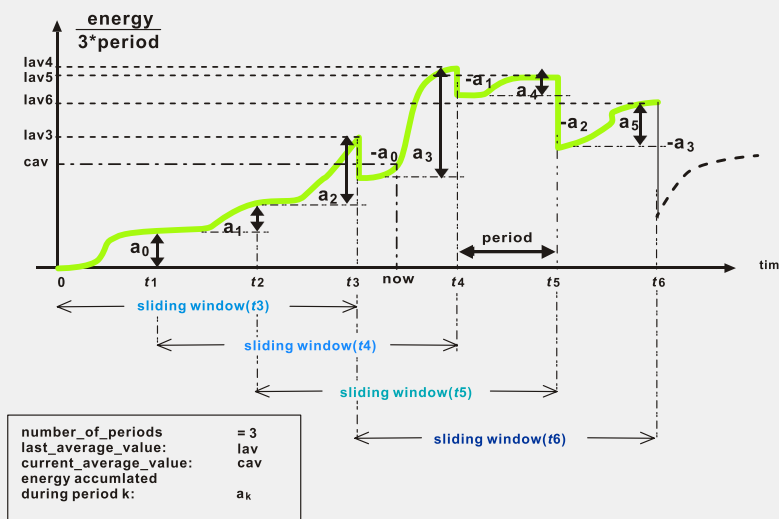
Accumulating MD: MD accumulating value of each month.

MD accumulation and calculation start after integral minute of initial power-on of meter.

- Block window: This is the demand calculation during a defined interval . Once the data is obtained, the value is stored and it makes a reset to start a new calculation for the next defined interval.



- Sliding window: This is the demand calculation during a defined interval. Once the data is obtained, it will wait one minute to start a new interval calculation (this time may vary depending on the customer). This means that every minute (this time can depends on the customer) it will record one maximum demand value from the last period.



- Current demand average value: (accumulating energy within current window/ numbers of period*period)
- Last demand average value: (accumulating energy within last window/ numbers of period*period)

3.10

Maximum Demand Registry

- Forward active MD register(+MA)

- Forward active tariff 1 MD register(+MA1)

- Forward active tariff 2 MD register(+MA2)

- Forward active tariff 3 MD register(+MA3)

- Forward active tariff 4 MD register(+MA4)

- Reverse active MD register(-MA)

- Reverse active tariff 1 MD register(-MA1)

- Reverse active tariff 2 MD register(-MA2)

- Reverse active tariff 3 MD register(-MA3)

- Reverse active tariff 4 MD register(-MA4)

- Forward reactive MD register(+MR)

- Forward reactive tariff 1 MD register(+MR1)

- Forward reactive tariff 2 MD register(+MR2)

- Forward reactive tariff 3 MD register(+MR3)

- Forward reactive tariff 4 MD register(+MR4)

- Reverse reactive tariff MD register(-MR)

- Reverse reactive tariff 1 MD register(-MR1)

- Reverse reactive tariff 2 MD register(-MR2)

- Reverse reactive tariff 3 MD register(-MR3)

- Reverse reactive tariff 4 MD register(-MR4)

- Forward apparent MD register(+MQ)

- Forward apparent tariff 1 MD register(+MQ1)

- Forward apparent tariff 2 MD register(+MQ2)

- Forward apparent tariff 3 MD register(+MQ3)

- Forward apparent tariff 4 MD register(+MQ4)

- Reverse apparent MD register(-MQ)

- Reverse apparent tariff 1 MD register(-MQ1)

- Reverse apparent tariff 2 MD register(-MQ2)

- Reverse apparent tariff 3 MD register(-MQ3)

- Reverse apparent tariff 4 MD register(-MQ4)

- Forward active historical MD register(+HMA)

- Forward active historical tariff 1 MD register(+HMA1)

- Forward active historical tariff 2 MD register(+HMA2)

- Forward active historical tariff 3 MD register(+HMA3)

- Forward active historical tariff 4 MD register(+HMA4)

- Forward active accumulation MD register(+CMA)

- Forward active accumulation tariff 1 MD register(+CMA1)

- Forward active accumulation tariff 2 MD register(+CMA2)

- Forward active accumulation tariff 3 MD register(+CMA3)

- Forward active accumulation tariff 4 MD register(+CMA4)

- Reverse active accumulation MD register(-CMA)

- Reverse active accumulation tariff 1 MD register(-CMA1)

- Reverse active accumulation tariff 2 MD register(-CMA2)

- Reverse active accumulation tariff 3 MD register(-CMA3)

- Reverse active accumulation tariff 4 MD register(-CMA4)

- Forward reactive accumulation MD register(+CMR)

- Forward reactive accumulation tariff 1 MD register(+CMR1)

- Forward reactive accumulation tariff 2 MD register(+CMR2)

- Forward reactive accumulation tariff 3 MD register(+CMR3)

- Forward reactive accumulation tariff 4 MD register(+CMR4)

- Reverse reactive accumulation MD register(-CMR)

- Reverse reactive accumulation tariff 1 MD register(-CMR1)

- Reverse reactive accumulation tariff 2 MD register(-CMR2)

- Reverse reactive accumulation tariff 3 MD register(-CMR3)

- Reverse reactive accumulation tariff 4 MD register(-CMR4)

- Forward apparent accumulation MD register(+CMQ)

- Forward apparent accumulation tariff 1 MD register(+CMQ1)

- Forward apparent accumulation tariff 2 MD register(+CMQ2)

- Forward apparent accumulation tariff 3 MD register(+CMQ3)

- Forward apparent accumulation tariff 4 MD register(+CMQ4)

3.11

Tariff Structure

Schedule Table

The tariff schedule contains exception day table, season table, week table and day table.

▪ Exception Day

Exception days table is formed by item, date, and corresponding day table ID. The meter compares the date with the entries in the exception days table. If the date is included in the table, it is an exception day. The season table and week table will be skipped, the day table ID will be directly used.

Exception days are divided to two classes: appointed exception days and public exception days.

Appointed exception days are valid in appointed years and public exceptions are valid in every year.

The exception day table can contain up to 100 entries.

▪ Season

The season table contains season name, valid beginning date and week name. Meter determines whether the current date is belong to the season table. If it is, the corresponding week table name will be obtained.

Just valid beginning time is available in the season table. If the current date is between two valid beginning dates, it will be judged as belong to the former season.

For example:

Spring 1st, March

Summer 1st, June

Autumn 1st, September

Winter 1st, December

If the current date is 1st, May, it will be judged belonging to Spring.

Up to 4 season tables could be set.

▪ Week

The week table defines the type of day valid in each case in lines for each day from Monday

to Sunday. The week table name and day table ID form the week table. The meter compares the current date with the corresponding week table and then gets the day table ID.

Up to 4 week tables could be set.

▪ Day

The day table contains time (hour and minute) and tariff Number. The meter compares the current time with the day table ID to get current tariff Number.

Up to 8 day table could be set.

The day table could contain up to 8 entries.

Calendar Switch Over

A passive calendar could be set with valid time. When the valid time comes, the passive calendar will be activated and replace the former calendar.

If the valid time of the passive calendar is before current time, the passive calendar will be activated immediately after being set. If the power grid is powered off leading the current time skips the valid time of passive calendar, the passive calendar will be activated immediately when power is on.

After the passive calendar is activated, the former passive calendar and activation time will be stored. But the mark of "To be activated" in meter will be cleared and the passive calendar will not be activated again.

An event of activation will be added to the event log after the activation of passive tariff table.

3.12

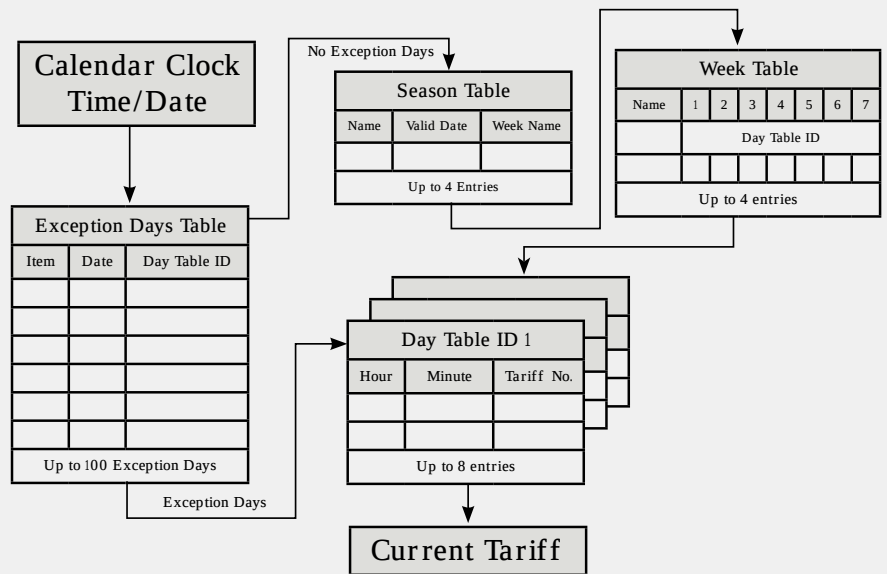
Types of Tariffs

TOU TARIFF SCHEDULE TABLE

The Time-of-use tariff structure includes day table, week table, season table and exception day table, while the back-up day table, back-up week table and back-up season table can be configured and activation time. When the back-up table is activated, it will replace the previous TOU schedule.

TOU SCHEDULES

- Tariff number: 4
- Time zone: 12
- Day timetable: 4
- Week table: 8
- Season table: 12
- Exception days: 100
- Default single tariff



STEP TARIFF

Step tariff is calculated based on the total kWh of the current month, and the tariff will be reset after the monthly billing date when the total kWh of the current month is reset. Backup step tariff table and activation time is configurable. After the backup tariff is activated, the backup tariff will replace the previous tariff table.

STEP KWH	TARIFF N°
STEP KWH 1	STEP TARIFF 1
...	...
STEP KWH 8	STEP TARIFF 8
THE METER SUPPORTS MAXIMUM STEP TARIFF 8	



Step tariff is not activated at default.

3.13

Tariff Table Schedule

TARIFF STRUCTURE CONFIGURATION

The tariff structure of the meter can be configured to ToU tariff, step tariff and mixed tariff. Backup tariff table and activation time is configurable. After the backup tariff is activated, the backup tariff will replace the previous tariff table.

TARIFF TABLE STRUCTURE

	TOU TARIFF 1	TOU TARIFF 2	TOU TARIFF 3	TOU TARIFF 4
STEP TARIFF 1	TARIFF 1_1	TARIFF 2_1	TARIFF 3_1	TARIFF 4_1
STEP TARIFF 2	TARIFF 1_2	TARIFF 2_2	TARIFF 3_2	TARIFF 4_2
STEP TARIFF 3	TARIFF 1_3	TARIFF 2_3	TARIFF 3_3	TARIFF 4_3
STEP TARIFF 4	TARIFF 1_4	TARIFF 2_4	TARIFF 3_4	TARIFF 4_4
STEP TARIFF 5	TARIFF 1_5	TARIFF 2_5	TARIFF 3_5	TARIFF 4_5
STEP TARIFF 6	TARIFF 1_6	TARIFF 2_6	TARIFF 3_6	TARIFF 4_6
STEP TARIFF 7	TARIFF 1_7	TARIFF 2_7	TARIFF 3_7	TARIFF 4_7
STEP TARIFF 8	TARIFF 1_8	TARIFF 2_8	TARIFF 3_8	TARIFF 4_8



1. Maximum 4 ToU tariff tables
2. Maximum 8 Step tariff tables
3. Maximum 4*8 mixed tariff tables
4. Default tariff are all 1.0000 Money/kWh

The meter deduct the credit based on the consumption of active energy.

CREDIT DEDUCT

- * **Single Tariff** - When consumption runs 0.01kWh, deduct remaining credit 0.01kWh
- * **ToU Tariff** - When consumption runs 0.01kWh, deduct remaining credit 0.01xToU kWh
- * **Step Tariff** - When consumption runs 0.01kWh, deduct remaining credit 0.01x Step kWh
- * **Mixed Tariff** - When consumption runs 0.01kWh, deduct remaining credit 0.01xMixed kWh



Credit deduction adopts single tariff mode at default.

3.14

Communication Ports

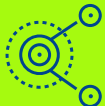


OPTICAL PORT



FEATURES

- Bidirectional serial communication, compliant with IEC 62056-21 standard
- Standby: 300bps. Meter communication after handshaking: 9600bps
- Mode E+HDLC communication mode



RS-485 PORT



FEATURES

- Bidirectional serial communication, compliant with IEC 62056-21 standard
- Communication baudrate: 9600bps
- HDLC communication mode



SCSSCAA9 PORT



PROTOCOL

- Bidirectional serial communication, compliant with IEC 62056-21 standard
- Standby: 300bps. Meter communication after handshaking: 9600bps
- Mode E+HDLC communication mode



EXCHANGEABLE MODEM



FEATURES

- Bidirectional serial communication
- Communication baudrate: 9600bps
- WRAPPER communication mode



COMMUNICATION SECURITY



FEATURES

Compliant with the standard DLMS/COSEM requirement, there are three levels of authorities for 3 clients:

- **Public client 0x10:** No security authentication visit. Data transfer uses plaintext. Only basic data objects can be read.
- **Read-only client 0x02:** Low security authentication visit. Data transfer uses plaintext. For meter reading only, meter relevant data can be read.
- **Management client 0x01:** High security authentication visit. Data transmission in cypher text. Meter data and configuration can be read and written.



FIRMWARE UPGRADE



LOCAL UPGRADE

- Local firmware upgrade adopts Y-Modem protocol. The meter must be kept powered on during the upgrade process. The meter runs the new firmware automatically after the upgrade.



IMAGE TRANSFER UPGRADE

- Image-transfer upgrade adopts Object 18 of DLMS/COSEM for upgrade and supports interruption resumption and the new firmware can be activated immediately after download or at defined time.

3.15

Relay Control

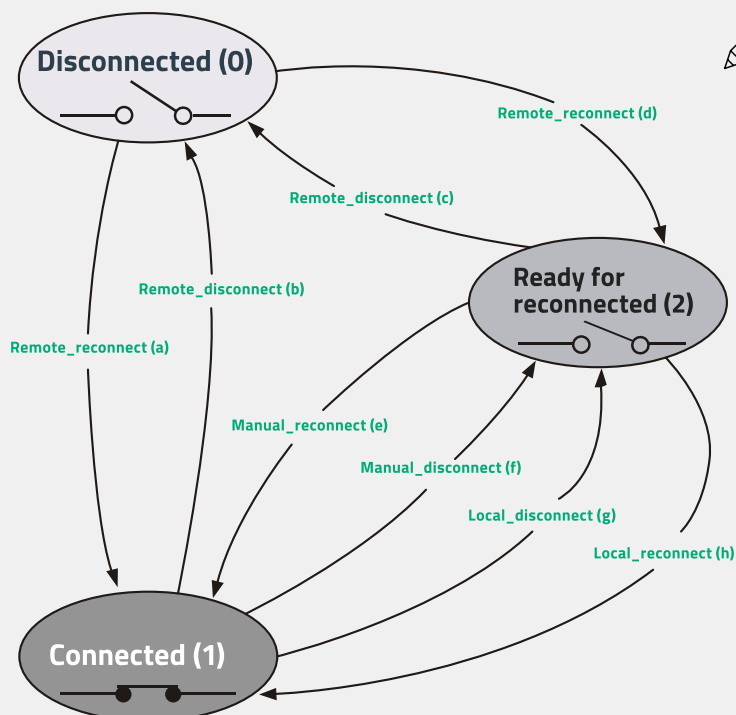
CONTROL STATUS

There are three statuses: disconnected (0), connected (1), ready for connected (2).

- Disconnected(0): relay is under disconnected status. Relay is not permitted to execute any operation before receiving connection order from Master Station/PC software or it is designated time for connection.
- Connected(1): relay is under connected status
- Ready for connected(2): relay is under disconnected status. Relay won't connect automatically.

CONTROL LOGIC

1. Local disconnect (g): When a disconnection event occurs, meter disconnects the supply. The relay is at status of ready for reconnect (2).
2. Local reconnect (h): When the meter recovers from disconnection event, meter connects the supply. The relay is at status of connect (1).
3. Remote disconnect (b): When the meter receives command remotely to disconnect, meter disconnects. The relay is at status of disconnect (0).
4. Remote reconnect (a): When receives command remotely to reconnect, meter reconnects supply. The relay is at status of connect (1).
5. Remote disconnect (c): When receives command remotely to disconnect, meter keeps disconnect. The relay is at status of disconnect (0).
6. Remote reconnect (d): When receives command remotely to reconnect, meter keeps disconnect. The relay is at status of ready for reconnect (2).
7. Manual disconnect (f): Input short code 999+ on the CIU (if available) to disconnect. The relay is at status of ready for reconnect (2).
8. Manual reconnect (e): Input short code 000+ on the CIU (if available) to reconnect. The relay is at status of connect (1).



When the relay is at status of ready for reconnect (2), the relay can be reconnected by

- Meter recover from disconnection event
- Input short code 000+ on the CIU (if available)
- Input clear tamper token

PHYSICAL FEATURES

10A

Maximum current control

100,000

Electrical cycle life

600A/10ms

Maximum short circuit current 50/60Hz 1 minute

100,000

Mechanical cycle life

100A 250VAC

Contact load

4000VAC

3.16

Mode of Relay Control

EXPLANATION

If current apparent power exceeds the valid overload threshold of (threshold active), the value of Overload (a register used to store delay of overload) will increase itself by 1 per second; if current apparent power less than Threshold active, Overload will decrease itself by 1 per second until 0. If the value of Overload exceeds period of delay which is configurable 1s~255s, an overload event will be recorded, triggering disconnection of relay.

After executing of relay disconnection (relay control status is Disconnected), the process of overload judgment won't be executed. After executing of relay connection, Overload will be reset and the process of over current judgment will start over.

EMERGENCY TIMETABLE MECHANISM

Meter can be configured with an emergency timetable, including the friendly hour set-up. If current time is defined in the emergency timetable, the meter will not disconnect at credit exhaustion. Relay disconnects automatically once the friendly hour finishes.

Relay will not reconnect for the same set-up once friendly hour mechanism has been used before the next recharge. Debit from the current friendly hour will be subtracted from any new credit entry.

Emergency timetable includes friendly hour, weekends, holidays. Emergency timetable is disabled at default.



MODE OF RELAY CONTROL

Control Mode	Relay Disconnect				Relay Reconnect			
	Remote		Manual	Local	Remote		Manual	Local
	(b)	(c)	(f)	(g)	(a)	(d)	(e)	(h)
(Mode_0)	✗	✗	✗	✗	✗	✗	✗	✗
(Mode_1)	✓	✓	✓	✓	✗	✓	✓	✗
(Mode_2)	✓	✓	✓	✓	✓	✗	✓	✗
(Mode_3)	✓	✓	✗	✓	✗	✓	✓	✗
(Mode_4)	✓	✓	✗	✓	✓	✗	✓	✗
(Mode_5)	✓	✓	✗	✓	✗	✗	✓	✓
(Mode_6)	✓	✓	✗	✓	✗	✗	✓	✓

 Default control mode of the relay is (Mode_6).

3.17

Load Control Mechanism

Program I Load Control

When the meter detects the consumed power is higher than the maximum load limit (Input short code **I3** and  on the CIU (if available) to check the overload limit value), the meter will activate the warning by flashing LED and the LCD displays **Over-P**. The overload lasts for 30 seconds, the meter will disconnect the power supply automatically.

After the power is disconnected, the supply will be resumed automatically 150 seconds after the overload disconnection. During the automatic reconnection, the LCD display the buffering countdown **EC-xxxx** (second). After overload disconnection occurs 4 times, the power supply can only be resumed after 30 minutes for the 5th disconnection.

Manual reconnection is available for the 5th disconnection by using **000** +  on the CIU (if available). After the 5th disconnection, the reconnection mechanism will restart from the 1st overload disconnection.

Key in digit **I3** and press  on the CIU (if available) to display the load limit of the meter.

Program II Load Limit Increase

If the meter disconnects the power supply constantly because of overload, the consumer can apply to the power utility to increase the maximum load threshold if applicable.

Relay Testing

Explanation:

- Input Token **0000 0000 0001 5099 7584** on the CIU (if available) to execute relay test.
- When the relay is on at default status, after the input of the test TOKEN, the relay will disconnect and the duration is 3 seconds. During the disconnection, it will not be affected by any command/event. After the disconnection finishes, the relay will reconnect automatically.
- When the relay is off at default status, after the input of the test TOKEN, the relay will reconnect and the duration is 3 seconds. During the reconnection, it will not be affected by any command/event. After the reconnection finishes, the relay will disconnect automatically.

Command Control

- **Reset Control:** Using the reset script to reset different data, including total reset (1), parameter reset (2), data reset (3), billing data reset (4), load profile reset (5), event reset (6) and tamper reset (7).
- **Command Billing Control:** Using the billing script to control the interface to make command billing remotely.
- **Remote Relay Control:** With relay control interface, it is possible to make the remote relay control on disconnection/reconnection. In the default relay control mode_6, remote disconnection command will disconnect the relay directly, and remote reconnection command will put the relay in status of "ready for reconnection". Key in digit **000** and press  on the CIU (if available) to reconnect the relay manually.

3.18

Firmware Upgrade

PROCESS OF UPGRADE

1

Initialization

PC software or central system initializes upgrading process and sends data packets for upgrading.

2

Checking

PC software or central system checks whether all packets have been correctly sent, if not, goes back to step (2), Software or central system resends packets that have not been correctly received.

5

Activation

Meter starts to update and active new version of software at the activation time. After upgrading, meter works with the new version of software.

3

Verification

PC software or central system reads version number of software to be upgrading and MD5 check code of packet. If they do not match, this upgrading fails, goes back to step (1), upgrading process starts over.

4

Setting

After confirmation of upgrading packet, PC software or central system sets activation time of upgrading packet. If the activation time is before current time, meter starts to active at once.

SAFETY PROTECTION OF UPGRADING

- When HHU or CAS starts the initialization of upgrading, meter checks whether the software version supports for upgrading. If not, upgrading will be rejected to prevent the use of incorrect upgrading packet by HHU or CAS.
- Each data frame is transmitted with CRC in compliance with DLMS protocol.
- Each mirror data block is with CRC.
- HHU or CAS checks if all data blocks have been successfully received.
- HHU or CAS checks software version ready for upgrading and MD5 check code stored in meter, in order to decide if the whole mirror upgrading packet has been successfully received.
- Meter performs another MD5 check on mirror upgrading packet before starting upgrading, and compares it with the previously generated MD5 check code.
- In order to ensure the correctness of each reading and writing of Image Code memory, the method of multi-comparison is adopted during the process of reading and writing.
- In case of wrong programming, it is checked again after programming.

3.19

Output Control Parameters

ALARM LED

Configuration

The alarm LED can be configured through control mode bit. The LED flashes at default when by-pass occurs. When the relay disconnects, the alarm LED stays on RED (top priority).

BUZZER

Configuration

The buzzer can be configured through control mode bit. The buzzer alerts at default when terminal cover open, meter cover open, chamber cover open, magnet tamper, inverse, over voltage, under voltage, over current and overload, no credit or low credit.

RELAY

Configuration

The Relay can be configured through control mode bit. The relay disconnects at default when terminal cover open, meter cover open, over voltage (any phase), under voltage (all phases), overload, and no credit. Clear tamper Tokens are required for tampers as terminal cover open and cover open.

Output Reason Status Bit

Bit	LED	BUZZER	RELAY
BIT0	Terminal cover open	Terminal cover open	Terminal cover open
BIT1	Meter cover open	Meter cover open	Meter cover open
BIT2	Meter box open	Meter box open	Meter box open
BIT3	Module cover open	Module cover open	Module cover open
BIT4	Magnet disturb	Magnet disturb	Magnet disturb
BIT5	Neutral disrupt	Neutral disrupt	Neutral disrupt
BIT6	High temperature	High temperature	High temperature
BIT7	Frequency anomaly	Frequency anomaly	Frequency anomaly
BIT8	Low power factor	Low power factor	Low power factor
BIT9	Overload	Overload	Overload
BIT10	Over voltage	Over voltage	Over voltage
BIT11	Under voltage	Under voltage	Under voltage
BIT12	Over current	Over current	Over current
BIT13	Inverse current	Inverse current	Inverse current
BIT14	Bypass	Bypass	Bypass
BIT15	Reserved	Reserved	Reserved
BIT16	Credit exhaustion	Credit exhaustion	Credit exhaustion
BIT17	Reserved	Low credit Level 1	Low credit Level 1
BIT18	Reserved	Low credit Level 2	Low credit Level 2
BIT19	Reserved	Low credit Level 3	Low credit Level 3
BIT 28	Reserved	Reserved	Under voltage in all phases
BIT29	Reserved	Reserved	Power outage disconnect
BIT30	Reserved	Reserved	Manual disconnect
BIT31	Reserved	Reserved	Remote disconnect

3.20

Real Time Clock (RTC)

Type

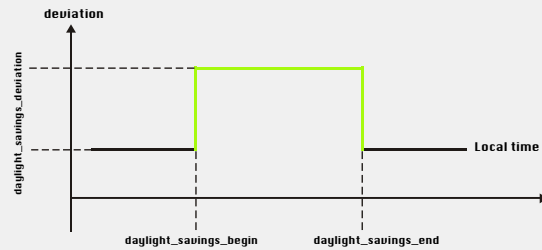
RTC consists of internal crystal oscillator, calendar clock and temperature compensation circuits. Every second, RTC sends an interruption signal to microcontroller and it interacts with microcontroller through I2C bus. Microcontroller is able to configure parameters of RTC and read calendar clock.

Parameter

- 23°C, deviation $\leq 1\text{s}$ per day.
- Within the range of normal temperature, the temperature drift $\leq 5\text{ppm}$, indicating that the deviation of the meter $\leq 0.15\text{s}$ per day.

Configuration

- Gregorian calendar
- Automatic leap year switch
- Configurable daylight saving time (DST starting and ending time for each year can be configured). The DST shift events will be added into the events log, the recorded time of events is in accordance with the shifted time. If DST mode is chosen, but the starting and ending time of DST (daylight saving time) is not configured, anytime is regarded as standard time.



- DST can be enabled or disabled, if meter is in the status of DST, at the same time, DST function is disabled, meter will automatically deducts one hour and switches into standard time. If meter is in the status of standard time and DST function is enabled, it will plus automatically one hour and switches into DST status. DST is disabled at default.

Provision

- Year (2000~2099)
- Month (01 ~ 12)
- Calendar day (01 ~ 31)
- Weekdays (01 ~ 07) 1...7, where 1=Monday, 2=Tuesday, etc.
- Hours (00 ~ 23)
- Minutes (00 ~ 59)
- Seconds (00 ~ 59)

Read-out

- The time of clock can be read and configured via communication ports and it should be configured with DST status, otherwise the meter might switch to DST automatically and a DST switch event would be added into the event log.

For example: the current time is:

Standard time: 13:00pm, July 24th, 2010

DST: 14:00pm, July 24th, 2010

The following situations that may occur when the clock is overwritten:

1, Written with: 14:00pm, 24th of July, 2010, DST.

The adjusted time will be: 14:00pm, 24th of July, 2010, DST. No event is recorded

2, Written with: 15:00pm, 24th of July, 2010, Standard time

The adjusted time will be: 15:00pm, 24th of July, 2010, DST. DST switch event is recorded.

3, Written with: 13:00pm, 24th of July, 2010, Standard time

The adjusted time will be: 14:00pm, 24th of July, 2010, DST. DST switch event is recorded.

4, Written with: 13:00pm, 24th of July, 2010, DST

The adjusted time will be: 13:00pm, 24th of July, 2010, DST. No event is recorded.

It's required to write the clock according the first method, because others may lead to mistakes.

3.21

Battery

Type

There is a lithium battery with 1250mAh in the meter, to maintain the calendar clock, record of opening meter cover and possible key-press wake-up display when power down. The battery is changeable and user authority is equal to plug in modem.

Voltage

The current battery voltage value is calculated via AD sample.

The sampling time points are:

- The power grid is power on again
- Every 10 seconds

Battery voltage Calculation accuracy of battery voltage is : $\pm 0.1V$

Normal battery voltage range: 3.4~3.8V

The battery voltage display value is not valid under power-down

Monitoring

The meter has the function of remaining battery power monitor.

The following formulas are used to calculate current remaining battery power :

- Current remaining battery power=(Remaining battery power before power down – Energy consumed by power down)
- Energy consumed by power down =Power down duration * Average power down working current

Average power down working current is the theoretical maximum average power down working current 40 μ A.

Life

NOTE

Remaining battery power is only for reference which is not standard calculated method.

Average working current

- Power grid is power on power on: $0.1\mu A < I < 1\mu A$
- Power grid is power down (displayed without pressing button) $15\mu A < I < 25\mu A$
- Power grid is power down (displayed with pressing button) $30\mu A < I < 60\mu A$

Changing Process

It can work for 15 years if total power grid is power down no more than 20000 hours.

The process of changing battery is as below:

- Changing hardware battery
- Executing the command of recovering remaining battery power
- Executing the command of clearing alarming register

After finishing the process of changing battery, the indicator of battery should be changed in broken-down register and the blinking flag of battery power displayed on LCD are cleared; operator of power supply company need to check whether the current battery voltage and remaining battery power are correct or not by button. If the battery is not changed correctly, the battery voltage is not in the normal range and the alarm of battery power shortage will be sent again within 10 seconds. If the battery is changed correctly, the battery voltage will recover to normal range.

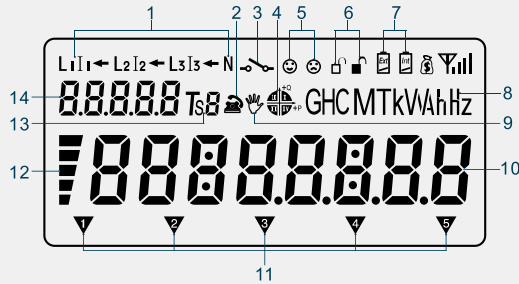
Note

If the hardware battery is changed when the power grid is power down, two cases should be paid attention to. In order to ensure no affect on the clock, the battery should be changed when the power grid is power on or reset the clock after the battery is changed.

Chapter 4 Display

Digital display is 8 bits (IEC 62056), visualized prompt symbol.
The type of LCD is alphanumeric 7-segment Liquid Crystal Display,
Digit dimension: 12mmX6.04mm,
Digit: 6 integers+2 decimals with OBIS code: 6 digits.

4.1 LCD Display Principle

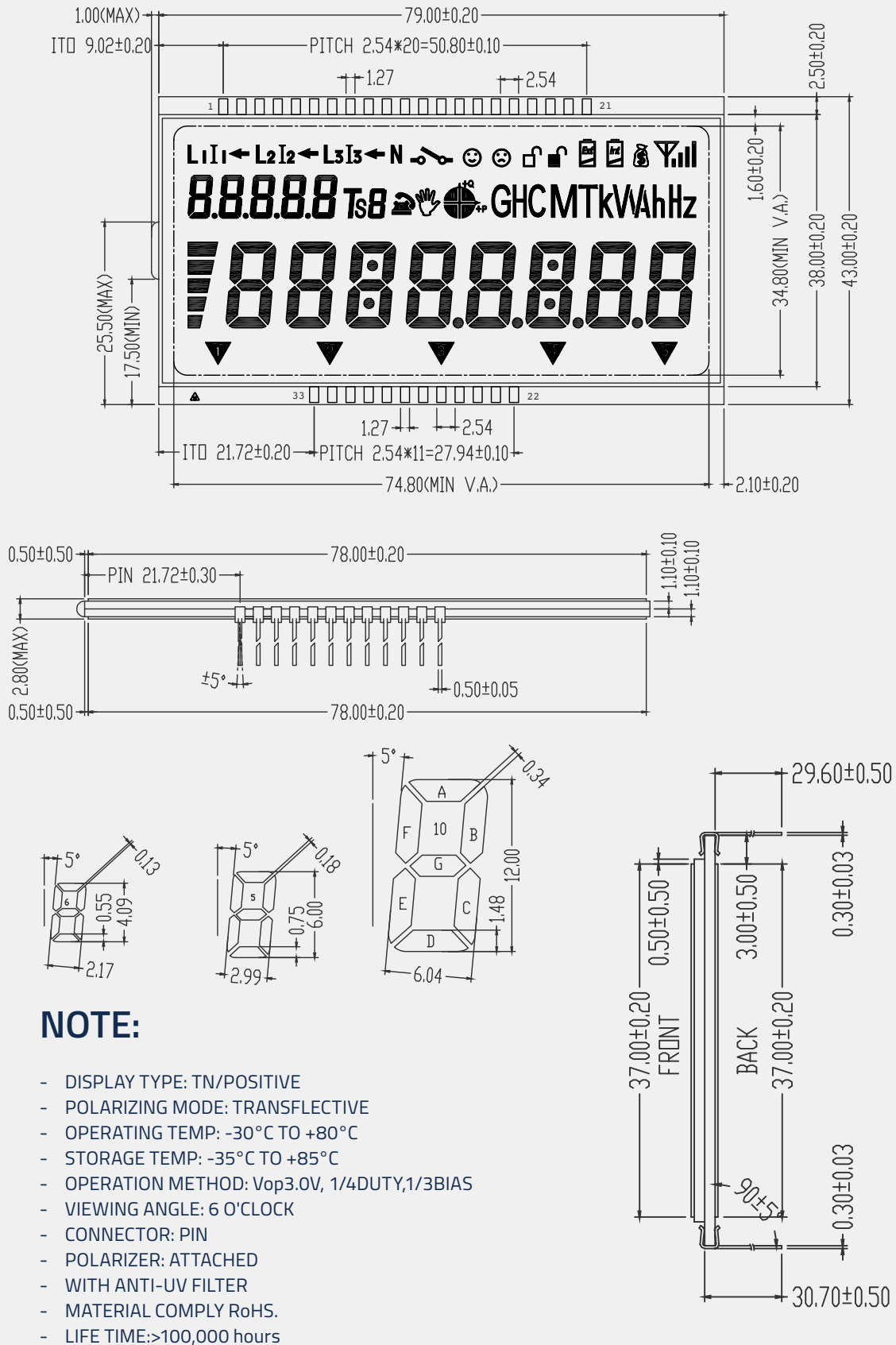


LCD // DISPLAY FORMATS

- Energy: 6 integers+2 decimals, in kWh/kVArh/kVAh
- Credit: 6 integers+2 decimals, in kWh
- Tariff: 4 integers + 4 decimals, in kWh
- Power/Demand: 4 integers+3 decimals, in kW/kAR/kVA
- Voltage: 3 integers+2 decimals, in V
- Current: 3 integers+3 decimals, in A
- Power factor: 1 integer+3 decimals, in PF
- Phase angle: 3 integers+1 decimal
- Frequency: 2 integers+2 decimals, in Hz

DISPLAY	EXPLANATION	DISPLAY	EXPLANATION
	DATA DISPLAY AREA METERING DATA	TkVAhHz	UNIT KWH: ACTIVE ENERGY KW: ACTIVE POWER KVARH: REACTIVE ENERGY KVAR: REACTIVE POWER KVA: APPARENT POWER V: VOLTAGE A: CURRENT HZ: FREQUENCY
	CODE DISPLAY AREA OBIS // SHORT CODE		
	CURRENT TARIFF RESERVED		CHEVRON (1) COMMUNICATION CHAMBER OPEN: FLASH (2) COMMUNICATION CHAMBER CLOSE: OFF
	REVERSED CONSUMPTION		CHEVRON (1) MAGNET DETECT: FLASH (2) MAGNET DISAPPEAR: OFF
	RELAY DISCONNECTION ON//RELAY DISCONNECT FLASH//READY FOR RECONNECT OFF//RELAY CONNECT		RTC ERROR CHEVRON
	TAMPER TAG		CHEVRON (1) POSTPAID MODE: ON (2) PREPAID MODE: OFF
	TOKEN ACCEPTANCE		CHEVRON (1) FACTORY MODE STEADY ON (2) NON-OPERATION MODE: FLASH (3) OPERATION MODE: OFF
	TOKEN REJECTION		BALANCE CREDIT INDICATING BAR WHEN ENERGY BALANCE <30 KWH, NO BAR DISPLAYS; WHEN 30 KWH <ENERGY BALANCE <60 KWH, THE FIRST BAR DISPLAYS; WHEN 60 KWH <ENERGY BALANCE <90 KWH, THE FIRST AND SECOND BAR DISPLAYS; WHEN 90 KWH <ENERGY BALANCE <120 KWH, THE 1ST, 2ND AND 3RD BAR DISPLAYS; WHEN 120 KWH <ENERGY BALANCE <150 KWH, THE 1ST, 2ND, 3RD AND 4TH BAR DISPLAYS; WHEN ENERGY BALANCE >150 KWH, ALL BAR DISPLAYS;
	QUADRANT INDICATOR NO-LOAD//COORDINATE ONLY WORK//QUADRANT INDICATION INDICATOR		
	METER COVER OPENING FLASH//OPENING UNDERGOING ON//OPENED WITHOUT CLEARING		
	TERMINAL COVER OPENING FLASH//OPENING UNDERGOING ON//OPENED WITHOUT CLEARING		GPRS SIGNAL INDICATOR SYMBOL 1) NO SYMBOL IS DISPLAYED WHEN THE MODULE IS NOT IN PLACE; 2) THE MODULE IS IN PLACE. TOWER SYMBOL FLASHES AND NO SIGNAL STRENGTH BAR IS DISPLAYED WHEN NO SIM CARD IS INSERTED OR MODULE IS INCORRECT. 3) THE MODULE IS IN PLACE AND THE SIGNAL TOWER SYMBOL FLASHES AND THE SIGNAL STRENGTH BAR FLASHES WHILE INITIALIZING; 4) THE MODULE IS IN PLACE, THE IP ACQUIRED, CONNECTING TO SERVER, TOWER SYMBOL ON, SIGNAL STRENGTH BAR FLASHES; 5) MODULE IN PLACE, ALREADY CONNECT THE SERVER, THE TOWER SYMBOL IS ALWAYS ON, AND THE SIGNAL STRENGTH BAR IS ALWAYS ON.
	INTERNAL BATTERY LOW VOLTAGE		
	EXTERNAL BATTERY LOW VOLTAGE		CURRENCY TAG RESERVED
GHC	GHANA CEDIS SYMBOL		COMMUNICATION TAG
L1 L2 L3	VOLTAGE INDICATOR FLASH//OVER VOLTAGE ON//VOLTAGE NORMAL OFF//UNDER VOLTAGE	I1 I2 I3	CURRENT INDICATOR FLASH//OVER CURRENT ON//CURRENT NORMAL OFF//CURRENT < 1ST

4.2 LCD Structural Details



4.3 Display Mode Explanation

POWER ON AUTO SCROLL DISPLAY

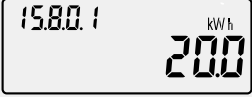
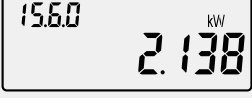
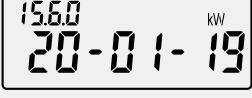
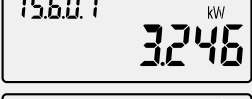
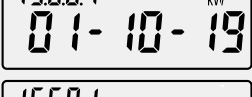
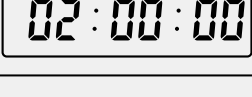
The meter will enter auto scroll display mode after it is energized. Each data item is displayed for 5 seconds. Press the push button to enter manual scroll mode. Long press push button to enter the lock screen display mode. If any event occurs, the event message will be displayed with the auto scroll display item alternately.

POWER ON MANUAL SCROLL DISPLAY

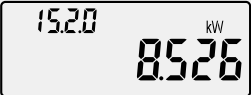
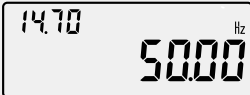
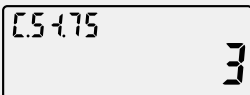
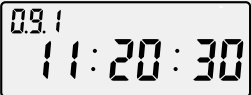
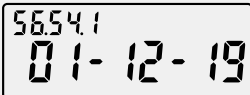
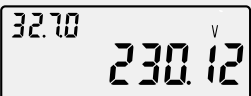
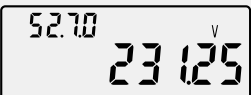
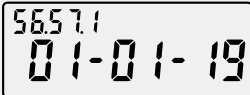
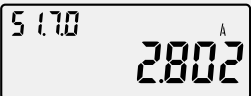
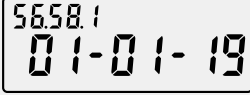
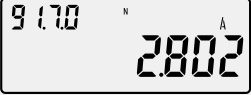
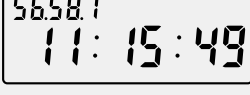
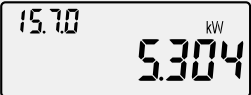
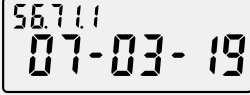
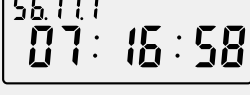
Press the push button to enter manual scroll mode. Press the button to scroll down and the data after entering the manual scroll mode. Long press the button to enter lock screen display mode, which will display the current data item. The meter will return to auto scroll display if no key is pressed for 30 seconds in the manual scroll mode.

Manual scroll display will be displayed in the sequence as on the right side and next page.

Long press the button to enter lock screen display mode, which will display the current data item. In lock screen display mode, the LCD backlit will stay on lighting and display the fixed data of the present display. Short press the push button to exit the lock screen mode.

DISPLAY	OBIS	EXPLANATION
	15.8.0	Total active energy
	C.50.0	Remaining credit
	15.8.0	Total active energy
	C.50.0	Remaining balance
	1.8.0	Import kWh
	2.8.0	Export kWh
	15.8.0.1	kWh of last month
	15.6.0	Active maximum demand of current month
		
		
	15.6.0.1	Active maximum demand of last month
		
		

4.3 Display Mode Explanation

DISPLAY	OBIS	EXPLANATION	DISPLAY	OBIS	EXPLANATION
	15.2.0	Cumulative maximum demand		14.7.0	Frequency
				C.1.0	Meter serial number
					
	0.9.2	Present date		C.54.75	Total tamper counts
	0.9.1	Present time		56.54.1	Last magnet disturbance occurrence date and time
	32.7.0	L1 voltage			
	52.7.0	L2 voltage		56.57.1	Last reverse consumptoin occurrence of L1 date and time
	31.7.0	L1 current			
	51.7.0	L2 current		56.58.1	Last reverse consumptoin occurrence of L2 date and time
	91.7.0	Neutral current			
	15.7.0	Combined active power		56.71.1	Last by-pass occurrence date and time
	13.7.0	Combined power factor			

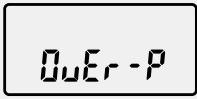
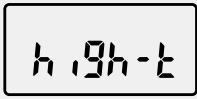
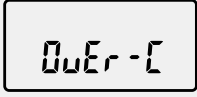
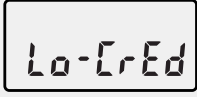
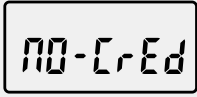
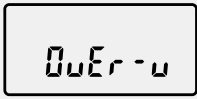
4.4 LCD Response Message to Token Input

TOKEN INPUT RESPONSE	DISPLAY	EXPLANATION
-	ACCEPT	TOKEN accepted by the meter
	REJECT	TOKEN rejected by the meter
	OLD	TOKEN validity expires
	USED	TOKEN has been used
	OK	Operation successful
	Err-01	Manufacture code error
	Err-02	Key expiration error
	Err-03	Key in meter has not been changed (DDTK type), refuse purchasing TOKEN
	Err-04	The energy is more than memory capacity
	Err-05	Key type for Changing the key TOKEN is not correct
	Err-06	TOKEN data format is not correct
	Err-07	TOKEN is not a type that can be accepted by the meter
	Err-08	The specified function in TOKEN does not exist
	LOCKED	After 5 wrong TOKEN input successively, TOKEN input is locked. Input 074 and press  to check the locked down time counting down (2 minutes at default)
	NULL	Not applicable

4.5 Event Display Explanation

EVENT DISPLAY EXPLANATION

- In auto-scroll display mode, when an event occurs, the event message will display with the scroll display items alternately. The event display can be configured, with default display including high temperature, terminal cover open, meter cover open, under voltage, over voltage, overload, low credit and no credit.

DESCRIPTION	DISPLAY	DESCRIPTION	DISPLAY
Over power		High temperature	
Over current		Low credit	
Meter cover open		No credit	
Terminal cover open		Over voltage	
Communication chamber cover open		Under voltage	

SETTLEMENT EVENT

- When billing settlement or demand settlement occurs, the LCD displays the settlement event for 3 seconds.

DESCRIPTION	DISPLAY	DESCRIPTION	DISPLAY
Billing Settlement		Demand Settlement	

DISPLAY AT POWER OFF

- The display can be waken up at meter power off by pressing the push button. After being waken up, the display will enter the mode of auto-scroll display at power off, and automatically shut off the display after a full display circle. If there is operation on the push button, the meter will enter display mode of manual scroll display at power off. The LCD display will shut off after the defined duration (30 seconds at default). Display items at power off is illustrated below.

DESCRIPTION	DISPLAY	DESCRIPTION	DISPLAY
Total kWh		Remaining credit	

Chapter 5 Meter Event Log

DS3100 detects and records various types of events and tampers and keeps its logs. It stores all registration and configuration data and event logs to a non-volatile memory. All data is retained for the life of the meter. Recordable security features are provided.

5.1 Types of Events

1. STANDARD EVENT

Time stamp and the event code. Maximum 100 entries.

#	EVENT	EVENT	#
1	Power outage occurs	Firmware downloaded	17
2	Power outage terminates	Firmware upgrade activated	18
3	Entry or exit of DST	ToU parameter configuration	19
4	Time before adjustment	Parameter changed	47
5	Time after adjustment	Global key changed	48
6	Invalid clock	Firmware authentication failed	51
8	Battery low voltage	Abnormal consumption/Relay anomaly	52
9	ToU activated	Reverse phase sequence	88
12	Firmware E2PROM error	Meter data reset	252
13	RAM Error	Demand reset	253
14	NVM Error	Load profile reset	254
15	Watchdog reset	Standard event reset	255
16	Measurement error	All events reset	256

2. TAMPER EVENT

Time stamp and the event code. Maximum 100 entries.

#	EVENT	EVENT	#
36	Module cover open	Inverse ends	107
37	Module cover close	L1 reverse power starts	200
38	Meter box open	L2 reverse power starts	201
40	Terminal cover open	L3 reverse power starts	202
41	Terminal cover close	L1 reverse power ends	203
42	Magnet tamper starts	L2 reverse power ends	204
43	Magnet tamper ends	L3 reverse power ends	205
44	Meter cover open	Bypass starts	206
45	Meter cover close	Bypass ends	207
49	Module cover open	Low power factor starts	208
50	Module cover close	Low power factor ends	209
91	Inverse starts	Tamper event reset	255

5.1

Types of Events

3. POWER QUALITY EVENT

Time stamp and the event code. Maximum 100 entries.

#	EVENT	EVENT	#
76	L1 under voltage starts	L2 over voltage ends	224
77	L2 under voltage starts	L3 over voltage ends	225
78	L3 under voltage starts	L1 voltage missing ends	226
79	L1 over voltage starts	L2 voltage missing ends	227
80	L2 over voltage starts	L3 voltage missing ends	228
81	L3 over voltage starts	L1 phase missing starts	229
82	L1 voltage missing starts	L2 phase missing starts	230
83	L2 voltage missing starts	L3 phase missing starts	231
84	L3 voltage missing starts	L1 phase missing ends	232
220	L1 under voltage ends	L2 phase missing ends	233
221	L2 under voltage ends	L3 phase missing ends	234
222	L3 under voltage ends	Voltage unbalance starts	235
223	L1 over voltage ends	Voltage unbalance ends	236
255	PQ event resets		

4. RELAY CONTROL EVENT

Time stamp and the event code. Maximum 100 entries.

#	EVENT	EVENT	#
59	Ready for reconnect	Remote reconnect	63
60	Manual disconnect	Local disconnect	64
61	Manual reconnect	Relay faulty	68
62	Remote disconnect	Local reconnect	69
255	Relay control event reset		

5. SELF-DEFINED EVENT

Time stamp and the event code. Maximum 100 entries.

#	EVENT	EVENT	#
240	L1 overload starts	L1 over current starts	246
241	L2 overload starts	L2 over current starts	247
242	L3 overload starts	L3 over current starts	248
243	L1 overload ends	L1 over current ends	249
244	L2 overload ends	L2 over current ends	250
245	L3 overload ends	L3 over current ends	251
255	Defined event reset		

5.1

Types of Events

6. COMMUNICATION EVENT

Time stamp and the event code. Maximum 100 entries.

#	EVENT	EVENT	#
140	Remote communication overtime	PDP context broke	147
141	Modem initialization failed	Received invalid PDP	148
142	SIM anomaly	Modem firmware reset	149
143	SIM returns functional	Modem hardware reset	150
144	GSM registration failed	User log-in failed	155
145	GPRS registration failed	Communication signal weak	156
146	PDP context established	Auto-response outnumber	157
255	Communication event reset		

7. CREDIT TOKEN

Time stamp and the event code, credit token, credit value, remaining credit and cumulative credit. Maximum 100 entries.

#	EVENT	EVENT	#
319	Credit token input	Credit TOKEN event reset	255

8. ENGINEERING TOKEN

Time stamp and the event code, engineering token, token type, cumulative kWh and cumulative credit, remaining credit. Maximum 100 entries.

#	EVENT	EVENT	#
320	Engineering token input	Engineering TOKEN event reset	255

Note: Type of Engineering Token

0x20	Load limit setting	Credit clearing	0x21
0x23	Key setting 1	Key setting 2	0x24
0x25	Tamper clearing	Key setting 3	0x28
0x3A	Extension setting		

5.2

Details of Events



01 //

TERMINAL COVER OPENING

-

RESTORATION :

1. Close the terminal cover
2. Wait till the LCD icon disappears
3. Input Clear Tamper Token (CTT)
4. Meter close the relay
5. Power supply is resumed
6. If input CTT without recovering the terminal cover, the meter will keep the alarm status without connecting the power supply.
7. In non-operation mode, the meter will only displays the meter cover indication, without cutting off the supply or recording the event.

OCCURRENCE :

When the terminal cover is open, and the duration is longer than the preconfigured delayed time, which is 1 second.

TERMINATION :

When the terminal cover is closed, and the duration is longer than the preconfigured delayed time, which is 5 seconds.

RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, alert LED lights up, audible buzzer alarms, LCD icon for the terminal cover  opening flashes and the relay icon  lights up, LCD displays **E - OPEN**. To save battery, the meter will only record the latest terminal cover opening even at power off.



SUPPORTS TERMINAL COVER OPEN DETECTION AT POWER OFF



02 //

METER COVER OPENING

-

RESTORATION :

1. Close the meter cover
2. Wait till the LCD icon disappears
3. Input Clear Tamper Token (CTT)
4. Meter close the relay
5. Power supply is resumed
6. If input CTT without recovering the meter cover, the meter will keep the alarm status without connecting the power supply.
7. In non-operation mode, the meter will only displays the meter cover indication, without cutting off the supply or recording the event.

OCCURRENCE :

When the meter cover is open, and the duration is longer than the preconfigured delayed time, which is 1 second.

TERMINATION :

When the meter cover is closed, and the duration is longer than the preconfigured delayed time, which is 5 seconds.

RESULT :

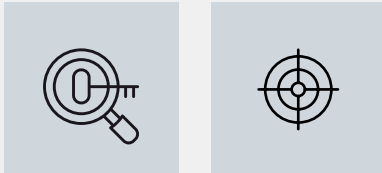
The meter records the occurrence date and time, the meter disconnects the power supply, alert LED lights up, audible buzzer alarms, LCD icon for the meter cover opening  flashes and the relay icon  lights up, LCD displays **E - OPEN**. To save battery, the meter will only record the latest cover opening even at power off.



SUPPORTS METER COVER OPEN DETECTION AT POWER OFF

5.2

Details of Events



ACTIONS :

1. Close the communication chamber and seal the chamber cover.
2. Wait till the LCD icon disappears
3. Input Clear Tamper Token (CTT)
4. Meter close the relay
5. Power supply is resumed
6. If input CTT without recovering the terminal cover, the meter will keep the alarm status without connecting the power supply.
7. The LCD icon ▼ goes off.

03 //

MODEM CHAMBER COVER OPEN

-

OCCURRENCE :

When the modem chamber cover is open, and the duration is longer than the preconfigured delayed time, which is 1 second.

TERMINATION :

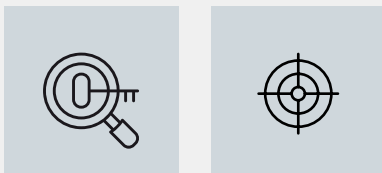
When the modem chamber cover is closed, and the duration is longer than the preconfigured delayed time, which is 5 seconds.

RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, the icon ▼ on LCD flashes, audible buzzer alarms, the relay icon ⚡ lights up, and the LCD displays **Ⓛ - OPEN**.



SUPPORTS METER COVER OPEN DETECTION AT POWER OFF



ACTIONS :

1. Close the communication chamber and seal the chamber cover.
2. Wait till the LCD icon disappears
3. Input Clear Tamper Token (CTT)
4. Meter close the relay
5. Power supply is resumed
6. If input CTT without recovering the terminal cover, the meter will keep the alarm status without connecting the power supply.

04 //

METER BOX COVER OPEN

-

OCCURRENCE :

When the meter box cover is open, and the duration is longer than the preconfigured delayed time, which is 1 second.

TERMINATION :

When the meter box cover is closed, and the duration is longer than the preconfigured delayed time, which is 5 seconds.

RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, audible buzzer alarms, the relay icon ⚡ lights up, and the LCD displays **Ⓛ - OPEN**.



SUPPORTS METER COVER OPEN DETECTION AT POWER OFF

5.2

Details of Events



05 //

MAGNET DISTURBANCE

ACTIONS :

1. Remove external magnet;
2. After the magnet is removed, the LED flashing stops, and LCD chevron stops .

OCCURRENCE :

When the meter detects external magnetic field disturbance and the duration exceeds the setting delayed time.

TERMINATION :

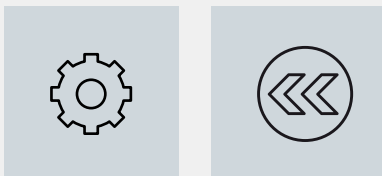
When the external magnetic field is removed, and the duration exceeds the setting delayed time.

RESULT :

The meter records the occurrence date and time, the warning LED flashes, and the LCD displays chevron of the magnet influence . The meter can be configured to display ~~0000-0000~~ (MAG-DST), which is not displayed at default.



MAGNET INFLUENCE SETTING DELAYED TIME: 30 SECONDS



06 //

L1 REVERSE CONSUMPTION

-

ACTIONS :

1. Correct the connection
2. Input the CTT
3. The relay closes, alarm LED goes off, reverse icon goes off and tamper tag goes off
4. The meter will not resume the power supply after the connection is corrected, but without inputting CTT

OCCURRENCE :

When the reverse power in L1 is higher than the preconfigured value for reverse power, and the duration exceeds the setting delayed time.

TERMINATION :

When the reverse power in L1 becomes lower than the preconfigured value, and the duration exceeds the setting delayed time.

REVERSE CONSUMPTION VALUE :

$$5\% \cdot U_A \cdot I_1$$

RESULT :

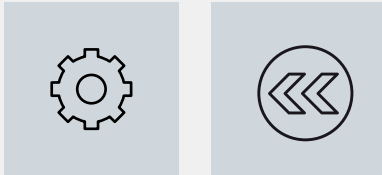
The meter records the occurrence date and time, the meter disconnects the power supply, LCD icon  for L1 reverse consumption lights up and the relay icon  lights up when applicable, alert LED lights up and the buzzer emits audible warning.



REVERSE CURRENT SETTING DELAYED TIME: 30 SECONDS

5.2

Details of Events



07 //

L2 REVERSE CONSUMPTION

-

ACTIONS :

1. Correct the connection
2. Input the CTT
3. The relay closes, alarm LED goes off, reverse icon goes off and tamper tag goes off
4. The meter will not resume the power supply after the connection is corrected, but without inputting CTT, the warning status remains

OCCURRENCE :

When the reverse power in L2 is higher than the preconfigured value for reverse power, and the duration exceeds the setting delayed time.

TERMINATION :

When the reverse power in L2 becomes lower than the preconfigured value, and the duration exceeds the setting delayed time.

REVERSE CONSUMPTION VALUE :

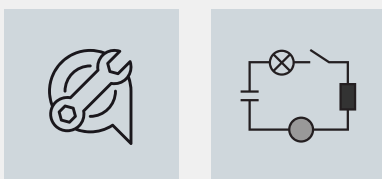
$$5\% \cdot U_B \cdot I_2$$

RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, LCD icon  for L2 reverse consumption lights up and the relay icon  lights up when applicable, alert LED lights up and the buzzer emits audible warning.



REVERSE CURRENT SETTING DELAYED TIME: 30 SECONDS



08 //

BY-PASS ($I_p > I_n$)

-

ACTIONS :

1. Check the wiring and correct the wiring.

OCCURRENCE :

When the vector current in phase or neutral is more than $5\%I_b$ and $(I_{max} - I_{min})/I_{max}$ exceeds the by-pass threshold (6.25% at default) and the duration exceeds the setting delayed time (30 seconds at default).

TERMINATION :

When the current is $I_1 + I_2 + I_3 = I_n$, and the duration exceeds the setting delayed time, or power off.

RESULT :

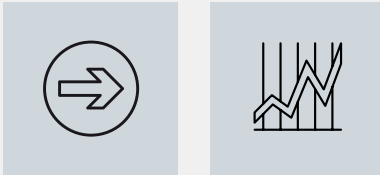
The meter can be configured to display **by-PASS** (not displayed at default), the meter records the occurrence date and time, alert LED lights up and the buzzer emits audible alarm.



UNBALANCED SETTING DELAYED TIME: 30 SECONDS
BA-PASS VALUE: 6.25%

5.2

Details of Events



09 //

OVER-CURRENT

-

ACTIONS :

1. Reduce consumption and wait till the buzzer silenced.

OCCURRENCE :

When the current exceeds the over-current threshold (I_{max} at default) and the duration exceeds the setting delayed time.

TERMINATION :

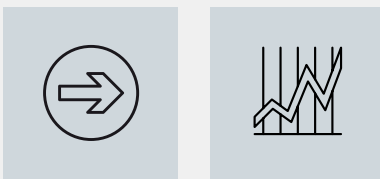
When the current becomes lower than the over-current threshold, and the duration exceeds the setting delayed time.

RESULT :

The meter records the occurrence date and time, the LCD can be configured to display **Over - C** (not displayed at default) and the buzzer emits audible warning.



OVER-CURRENT SETTING DELAYED TIME: 30 SECONDS
OVER-CURRENT THRESHOLD: I_{max}



10 //

OVER-CURRENT

-

ACTIONS :

1. Reduce consumption and wait till the buzzer silenced.

OCCURRENCE :

When the current exceeds the over-current threshold (I_{max} at default) and the duration exceeds the setting delayed time.

TERMINATION :

When the current becomes lower than the over-current threshold, and the duration exceeds the setting delayed time.

RESULT :

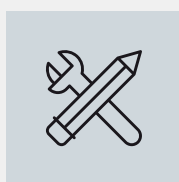
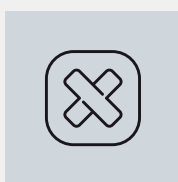
The meter records the occurrence date and time, the LCD can be configured to display **Over - C** (not displayed at default) and the buzzer emits audible warning.



OVER-CURRENT SETTING DELAYED TIME: 30 SECONDS
OVER-CURRENT THRESHOLD: I_{max}

5.2

Details of Events



11 //

POWER OUTAGE

-

ACTIONS :

1. The meter records the power outage date and time;
2. The meter records the power resume date and time;

OCCURRENCE :

When voltage of any phase drops below the starting voltage of the meter.

TERMINATION :

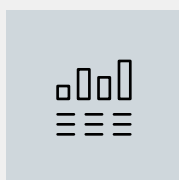
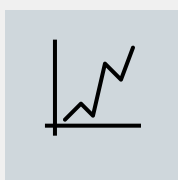
When the voltage resumes to above the starting range.

RESULT :

The meter records the occurrence date and time.



RECORD SETTING DELAYED TIME: IMMEDIATELY



12 //

OVERLOAD

-

ACTIONS :

1. Wait for 150 seconds for 1st to 4th time;
2. For 5th time, wait 30 minutes;

OCCURRENCE :

When the instantaneous power measured by the meter is higher than the permitted power load ($I_{max} \times U_n$ at default) and lasts for more than 30 seconds, the meter will disconnect the power supply. When disconnection occurs due to overload, the meter can be reconnected by pressing any key on the CIU. If the meter trips 5 times continuously, the meter will only resume the power supply after 30 minutes at the power disconnection for the 5th time, and the power can not be resumed manually.

RESULT :

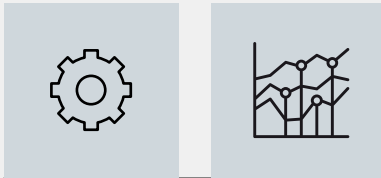
The meter records the occurrence date and time, the meter disconnects the power supply, LCD displays **Over-P**, the relay icon  lights up, the warning LED flashes, and the buzzer emits audible warning.



OVERLOAD SETTING DELAYED TIME: 30 SECONDS

5.2

Details of Events



ACTIONS :

1. Auto restoration: When the voltage becomes below $120\%U_n$ and the duration exceeds 5 minutes;
2. Manual restoration: When the voltage becomes below $120\%U_n$ and press the push button on the meter;

13 //

L1 OVER-VOLTAGE

-

OCCURRENCE :

When the L1 voltage exceeds $120\%U_n$ and the duration exceeds 30 seconds, LCD can be configured to display $\overline{U_{L1}}$ (not display at default), the buzzer alarms and the relay disconnects at over-voltage of any phase at default.

TERMINATION :

When the L1 voltage becomes below $120\%U_n$ and the duration exceeds 5 minutes. Or press push button the meter after the voltage returns below $120\%U_n$

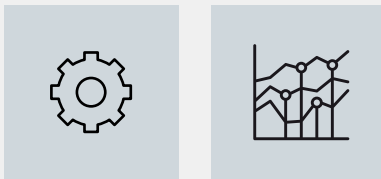
RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, the relay icon  lights up, the warning LED flashes and the buzzer emits audible warning. The LCD displays the countdown for relay reconnection.

When the countdown for reconnection is 0, the meter will automatically reconnects the power supply.



OVER-VOLTAGE SETTING DELAYED TIME: 30 SECONDS
OVER-VOLTAGE LIMIT: DEFAULT $120\%U_n$



ACTIONS :

1. Auto restoration: When the voltage becomes below $120\%U_n$ and the duration exceeds 5 minutes;
2. Manual restoration: When the voltage becomes below $120\%U_n$ and press the push button on the meter;

14 //

L2 OVER-VOLTAGE

-

OCCURRENCE :

When the L2 voltage exceeds $120\%U_n$ and the duration exceeds 30 seconds, LCD can be configured to display $\overline{U_{L2}}$ (not display at default), the buzzer alarms and the relay disconnects at over-voltage of any phase at default.

TERMINATION :

When the L2 voltage becomes below $120\%U_n$ and the duration exceeds 5 minutes. Or press push button the meter after the voltage returns below $120\%U_n$

RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, the relay icon  lights up, the warning LED flashes and the buzzer emits audible warning. The LCD displays the countdown for relay reconnection.

When the countdown for reconnection is 0, the meter will automatically reconnects the power supply.



OVER-VOLTAGE SETTING DELAYED TIME: 30 SECONDS
OVER-VOLTAGE LIMIT: DEFAULT $120\%U_n$

5.2

Details of Events



15 //

L1 UNDER-VOLTAGE

-

ACTIONS :

1. Auto restoration: When the voltage becomes above $60\%U_n$ and the duration exceeds 5 minutes;
2. Manual restoration: When the voltage reaches above $60\%U_n$ and press push button on the meter;

OCCURRENCE :

When the L1 voltage drops below $60\%U_n$ and the duration exceeds 30 seconds, and LCD displays **L1-U** (not display at default), the buzzer alarms and the relay disconnects at under-voltage of any phase at default;

TERMINATION :

When the L1 voltage becomes above $60\%U_n$ and the duration exceeds 5 minutes.
Or press the push button on the meter after the voltage returns above $60\%U_n$

RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, the relay icon  lights up, the warning LED flashes and the buzzer emits audible warning. The LCD displays the countdown for relay reconnection. When the countdown for reconnection is 0, the meter will automatically reconnects the power supply.



UNDER-VOLTAGE SETTING DELAYED TIME: 30 SECONDS
UNDER-VOLTAGE LIMIT: DEFAULT $60\%U_n$



16 //

L2 UNDER-VOLTAGE

-

ACTIONS :

1. Auto restoration: When the voltage becomes above $60\%U_n$ and the duration exceeds 5 minutes;
2. Manual restoration: When the voltage reaches above $60\%U_n$ and press push button on the meter;

OCCURRENCE :

When the L2 voltage drops below $60\%U_n$ and the duration exceeds 30 seconds, and LCD displays **L2-U** (not display at default), the buzzer alarms and the relay disconnects at under-voltage of any phase at default;

TERMINATION :

When the L2 voltage becomes above $60\%U_n$ and the duration exceeds 5 minutes.
Or press the push button on the meter after the voltage returns above $60\%U_n$

RESULT :

The meter records the occurrence date and time, the meter disconnects the power supply, the relay icon  lights up, the warning LED flashes and the buzzer emits audible warning. The LCD displays the countdown for relay reconnection. When the countdown for reconnection is 0, the meter will automatically reconnects the power supply.



UNDER-VOLTAGE SETTING DELAYED TIME: 30 SECONDS
UNDER-VOLTAGE LIMIT: DEFAULT $60\%U_n$

5.2

Details of Events



17 //

RELAY ON/OFF

-

ACTIONS :

1. Relay disconnects/connects power supply
2. Meter records occurrence date and time.

OCCURRENCE :

Relay trips when meter disconnection condition occurs.

TERMINATION :

Meter restores from disconnection condition and/or input CTT.

CONDITION :

Relay tripping: Terminal cover/meter cover open, reversed, overload, over-voltage, under-voltage

RESULT :

The meter records the occurrence date and time of relay trips and connection of power supply.



RECORD SETTING DELAYED TIME: IMMEDIATELY



18 //

RECHARGE

-

ACTIONS :

1. Input the Recharge Credit TOKEN into the meter;

OCCURRENCE :

Credit recharge

RESULT :

The meter records the occurrence date and time, the recharge credit amount and TOKEN.



RECORD SETTING DELAYED TIME: IMMEDIATELY



19 //

MEMORY ERROR

-

ACTIONS :

1. Replace the meter;

OCCURRENCE :

When there is operation error in Non Volatile Memory (E2PROM, DATA FLASH).

RESULT :

The meter records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY

5.2

Details of Events



20 //

ENGINEERING TOKEN

-

ACTIONS :

1. Input the Engineering TOKEN into the meter;

OCCURRENCE :

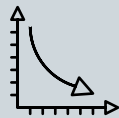
Meter input engineering token

RESULT :

The meter records the occurrence date and time, the type of token and TOKEN.



RECORD SETTING DELAYED TIME: IMMEDIATELY



21 //

LOW POWER FACTOR

-

ACTIONS :

1. Records and stores the occurrence date/time history and the event regardless of the returning point;

OCCURRENCE :

Low power factor indicates when the PF within a certain set period (30 seconds at default) is lower than the setting range (0.8 at default).

CONDITION :

The detection is made on a certain threshold (initial 0.8 at default), and the current is over $5\%I_b$, while also it lasts more than a certain duration (initial 30 seconds at default).

RESULT :

Records and stores the occurrence date/time history and the event regardless of the returning point.



RECORD SETTING DELAYED TIME: IMMEDIATELY



22 //

CLOCK ERROR

-

ACTIONS :

1. Adjust the clock;

OCCURRENCE :

When the clock value of the meter is not correct.

RESULT :

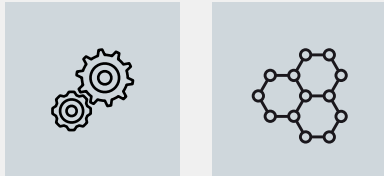
The meter displays  icon calling for clock adjustment and records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY

5.2

Details of Events



23 //

INTERNAL BATTERY LOW VOLTAGE

-

ACTIONS :

1. Replace the internal battery if applicable;

OCCURRENCE :

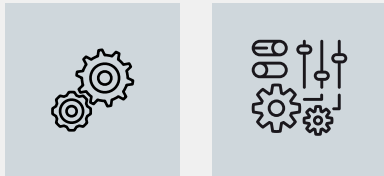
When the internal battery voltage is lower than the threshold and lasts longer than the defined duration (30 seconds at default).

RESULT :

The meter displays the internal battery low voltage icon  and records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY



24 //

EXTERNAL BATTERY LOW VOLTAGE

-

ACTIONS :

1. Replace the external battery;

OCCURRENCE :

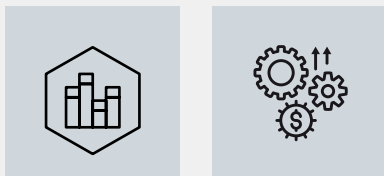
When the external battery voltage is lower than the threshold and lasts longer than the defined duration (30 seconds at default).

RESULT :

The meter displays the external battery low voltage icon  and records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY



25 //

INTERNAL HIGH TEMPERATURE

-

ACTIONS :

1. Report to your energy supplier or reduce the consumption to cool off the meter;

OCCURRENCE :

When the internal temperature is higher than the threshold and lasts longer than the defined duration (30 seconds at default).

RESULT :

The meter displays  and records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY

5.2

Details of Events



26 //

WATCHDOG RESET

-

ACTIONS :

1. No action is called for;

OCCURRENCE :

When the meter recovers from error and reset, the meter will consider it as watchdog reset.

RESULT :

The meter records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY



27 //

RELAY FAULTY

-

ACTIONS :

1. Call for meter replacement;

OCCURRENCE :

When the meter detects current after the relay trips, it will be recorded as relay faulty.

RESULT :

The meter records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY



28 //

MEASUREMENT ERROR

-

ACTIONS :

1. Call for meter replacement;

OCCURRENCE :

When the registers are not correct, the meter will consider it as measurement error.

RESULT :

The meter records the event.



RECORD SETTING DELAYED TIME: IMMEDIATELY

5.3

Self-Diagnosis of Faulty, Alarm and Status

- ✓ The meter stores data and detect itself every day and when the power is on. The meter also monitors the operation status real time and display the operation status in LCD. The master station (data center) obtains the alarm, AMI status to determine whether the meter performs normally.



FAULTY REGISTER

Replace battery only can be cleared by communication. The other will be cleared automatic when error disappear. When occurrence of NV memory error, When the NV memory has fault, the meter will display the information for indicating the deadly faults occurrence.



ALARM REGISTER

The meter is displaying if clock is at error, the physical status of the contactor is different from the required status, data error such as measurement system error, watchdog error, fraud attempt.



AMI STATUS

The meter detects Clock reset exceeds the limit, Calibration has error, the register is faulty or meter is in ToU mode error. The statuses only could be reset and they could not be cleared automatically except Daylight saving indication. Only the master station and PC has the authorization to clear these statuses.



Advanced features as meter pushing alarms to the Master Station (Data Center) is available when the meter deployed in a AMI environment.

Data Concentrator Unit also pulls status information from the meter then the meter is deployed with DCU, which manages the meter status for settlement.



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